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Review

Biochar for ameliorating soil fertility and microbial diversity: From production to action of the black gold

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SUMMARY

This article evaluated different production strategies, characteristics, and applications of biochar for ameliorating soil fertility and microbial diversity. The biochar production techniques are evolving, indicating that newer methods (including hydrothermal and retort carbonization) operate with minimum temperatures, yet resulting in high yields with significant improvements in different properties, including heating value, oxygen functionality, and carbon content, compared to the traditional methods. It has been found that the temperature, feedstock type, and moisture content play critical roles in the fabrication process. The alkaline nature of biochar is attributed to surface functional groups and addresses soil acidity issues. The porous structure and oxygen-containing functional groups contribute to soil microbial adhesion, affecting soil health and nutrient availability, improving plant root morphology, photosynthetic pigments, enzyme activities, and growth even under salinity stress conditions. The review underscores the potential of biochar to address diverse agricultural challenges, emphasizing the need for further research and application-specific considerations.

INTRODUCTION

Globally, there is a notable increase in commodities consumption which is attributable to the globe's dedication to improved economy and living conditions. This notable rise in commodities consumption has directly influenced the quantity of agro-waste generated, about a hundred million tons of agro-waste is generated globally,[1,2] for analogy: Agriculture is globally identified as a major and significant sector of the economy, providing employment for over 60 percent of the world population while contributing to over 3.5% of the GDP ca. 2020, while accounting for about $ 3.2 trillion of the global GDP ($100 billion of the Canada's GDP), Agricultural processes also contribute to over 130 billion tons of the global waste (>27 million tons of the waste in Canada).[3] India, an agro-based country generating over 30 percent of its GDP from agricultural practice experiences over 600 million tons of biowaste annually where about 20% of Indian fruits end up as agro-wastes.[1–3] These agricultural wastes pose hazards to environmental sustainability when improperly treated or left untreated. Environmental concerns such as a significant increase in Green House Gases (GHG) are directly linked to untreated or not adequately treated agro-wastes.[4] Biomass from agro and municipal solid wastes are one of the major promoters of GHGs, where gases such as CO_2, CH_4 and N_2O excreted from agriculture contribute to over 10–13% of the world's and Canada's GHG emissions respectively.[5,6] These biomasses are composed of bio-chemically active components, which makes them alternative sources of materials for different applications when thermally converted employing techniques such as pyrolysis, gasification, and direct combustion.[7]

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Biochar production has been identified as the most sustainable waste management technique due to the defectiveness of available waste management methods in most third worlds, and the rise in satisfactory waste disposal costs.[8] Biochar, a carbon-rich charred product obtained from the thermo-chemical conversion of biomass via the pyrolysis technique (conversion in the absence of oxygen), has gained more interest over the years as a result of its distinctive properties such as significant cation exchange capacity, high porosity with large surface area and good adsorbent which makes it suitable for a wide range of environmental application such as wastewater treatment, carbon sequestration, and soil remediation.[9–11] Huggins et al.[12] compared the performance of biomass-derived char to activated carbon for wastewater treatment and preconcentration of its nutrients, biochar displayed greater nutrient recovery and adsorption capacity owing to its macro pores compared to the activated carbon micropores, results from this study and biochar reusability owing to it life cycle has made it a sustainable material for packed bed wastewater treatment. Although farmers use fertilizer for soil amendment which directly increases the yield of the farm produce by improving the biological, physical, and chemical characteristics of the soil via the release of essential nutrients such as phosphorus, potassium nitrogen, and others based on the fertilizer properties to the soil.[13] Unfortunately, the soil does not completely utilize the nutrients released by these fertilizers, thus resulting in nutrient accumulation and release into the environment.

Biochar has been described as the "black gold" in agricultural circles owing to its exceptional features such as high composition of stable carbon. This aids long-term carbon immobilization in soil and improves fertility due to its controlled release of locked up carbon and nutrients for several decades. This also helps to ameliorate climate change due to reduced atmospheric carbon and gases from recurring fertilizer applications. Also, biochar's large surface area and porosity aid water and nutrient retention alongside beneficial microbes' growth in soil directly increasing agricultural yield by improving the soil ecosystem while optimizing productivity through the neutralization of soil acidity which provides a balanced environment, thereby promoting crop development.[14,15] All these exceptional features which made biochar a sustainable and efficient means for long term carbon sequestration, soil fertilizing, and amendment are equated to that of precious metals such as gold, thus fostering the tagging of biochar as "black gold".

Biochar for soil amendment could be traced back to the investigation of Glaser and Birk[16] on Amazonian soils; a comparison of the crop yields from Amazonian soils amended with biochar with normal Amazonian soils proved the application of carbonaceous material as a soil fertilizer. Restoration of lost soil nutrients to improve crop yields is the major driving force for the farmers' application of this "black gold" to soils.[17] Zhao et al.[18] compared the rate of nutrient leaching from inorganic fertilizers to biochar. Using NPK as the source of nutrients from inorganic fertilizer and coconut shell-derived biochar to evaluate the nutrient loss and yield of spinach via improved soil properties. Results from the tropical field experiment reveal less nitrogen loss from biochar treatment compared to NPK treatment. Alongside over 30% increase in spinach yield under biochar treatment compared to NPK treatment.

Additionally, traced metals availability and mobility in soil increases as the soil pH falls, owing to excess the application of inorganic fertilizers which results in nutrient accumulation and leaching. To mitigate these effects the most sustainable method to improve soil properties without releasing heavy metals to the environment is the use of modified biochar in place of inorganic or even organic fertilizer.[13,19] The major and common source of organic fertilizer is livestock waste. This waste and other wastes such as sewage sludge cannot be applied as a source of nutrients on soil due to the presence of unwanted micro-organisms which pose a huge risk factor, and require complete treatment and modification by mixing with biochar before application.[20] Furthermore, biochar's ability to improve soil water and nutrient holding capacity has increased its consideration for soil amendment, as it directly deals with environmental issues such as reduced water stress and emission of greenhouse gases via the provision of water to plants, continuing carbon sink while improving soil drainage, infiltration and availability of nutrient by modifying the soil physiochemical properties.[21,22] However, the influence of pyrolysis conditions and biomass type in biochar efficacy in soil remediation is not negligible. Wilson et al.[23] evaluated the effect of pyrolysis condition biochar performance on soil amendment, using a pyrolysis temperature between 400°C and 500°C, Biochar generated with calcium oxide is more stable and durable in soil.

Recently, some studies have been used to investigate the effect of pyrolysis parameters, biomass source, carbon concentration, and application concentration on the soil physiochemical characteristics.[24,25] Some studies were carried out to evaluate the effect of soil texture, biochar particle size, and biochar aging on soil amendment alongside considering the influence of biochar porosity and active functional groups.[26–28] This review aims to evaluate the performance of different biochar applied in soil amendment considering the biochar source and production parameters. The review puts all these together by comprehensively examining different biochar preparation strategies and influencing process factors on the properties. We discuss some of the key biochar characteristics and then delve into their physical and chemical influence on the soil. The review then discusses biochar's (i) relationship with soil microbial fauna and (ii) interaction with plants. This review was developed to serve as a bridge between biochar production experts and research in the agricultural application of biochar. We have presented in the same work a comprehensive analysis of ways in which biochar has been produced, systematically connected this with their observed properties and then showed how these properties are exploited or harnessed in soil science. Firstly, this work serves as a resource for biochar production experts to know what properties and characteristics are of interest to soil scientists. Secondly, it also serves as an analysis of the field to help us develop insight into the areas needing more research interest and where breakthroughs are more likely.

DIFFERENT APPROACHES TO SOIL REJUVENATION

Conservation of soil health and crop productivity is the central theme for sustainable agriculture practices. It is unrealistic to expect that the burgeoning crop production demands will be

met by a soil ecosystem that is increasingly unhealthy and constrained. To maintain soil health, minimize chemicals (coming from chemical fertilizers and pesticides), and maintain soil microbial diversity, it is necessary to rejuvenate soil from time to time. There are several biogenic products by which soil rejuvenation is practiced, including organic manures, composting, and biochar.

Rejuvenating soil health using organic manures

Organic manures are one of the effective materials that contain nutrients in complex organic forms used for soil rejuvenation, which involves natural, biologically active materials to improve soil fertility, structure, and overall health.[29] The organic manures, obtained from animals and residues contribute several carbonaceous matters, which act as a base exchange and material that provides mineral nutrients to the plants.[30] Organic manures, such as compost, green manure, animal manure, and other organic materials, help enhance soil quality by increasing nutrient content, improving water retention, fostering beneficial microbial activity, and reducing chemical inputs. It provides energy to soil microbes resulting in rich microbial diversity and biological activity in the soil. During decomposition, it produces some organic acid, which might help to dissolve soil minerals, including N, P, and K, to make them available to the plants for enhancing growth and crop yield.[29] One of the most important aspects of manure use is avoiding chemical fertilizers. Generally, two types of manures are used in the agricultural field: (a) concentrated organic manures and (b) bulky organic manures. Poultry manure is one of the nitrogen- and phosphorus-rich bulky manures, which ferment rapidly (>50% N lost within 30 days) if left exposed in the field. It contains ~13 essential plant-required nutrients. Application of poultry manure results in higher N, P, K, Cu, Mn, and Fe, which enhances the growth and yield of crops.[31]

When organic manures are being applied to the soil, they release these nutrients and organic matter slowly over time through microbial decomposition, making them available to plants. Organic matter acts as a binder, forming aggregates that create spaces for air and water movement within the soil. Generally, the microorganisms break down the organic matter, release nutrients, and produce substances that improve soil structure through nutrient cycling. The organic matter in manures can also improve soil aeration and water infiltration, reducing the risk of compaction, allowing it to retain moisture for longer periods, and waterlogging. A well-structured soil is beneficial for root development and enhances the overall resilience of the soil against erosion. The increased nutrient content leads to healthier plant growth, better crop yields, and improved overall soil fertility. Table 1 represents the effects of manure on physicochemical properties and fertility of the soil. Where it has been observed that most of the studies have been conducted in China the manure overall increased the available nutrient (NPK) of the soil.

To effectively rejuvenate soil health using organic manures, it is essential to choose appropriate manure types, apply them in the right quantities, and manage their decomposition to optimize the benefits while minimizing potential risks, such as nutrient imbalances or excessive nutrient leaching. Crop rotation and cover cropping practices can also complement the use of organic manures to further improve soil health and sustainable agriculture.

Rejuvenating soil health using compost

Composting is a natural and effective way to rejuvenate soil health. It involves the controlled decomposition of organic materials, such as kitchen scraps, yard waste, agricultural residues, and animal manure, to produce a nutrient-rich compost. It is a self-heating, thermophilic, aerobic process of organic matter decomposed by microorganisms under controlled conditions. Composting is basically the conversion (oxidation of organic carbon) of plant and animal-derived organic wastes into nutrient-enriched biomass through decomposition using bacteria, actinobacteria, protozoa, and fungi.[29] The main advantages of composting are enhancing the available nutrients (macro and micro) for plant growth without harming the environment. It is an N, P, and K-rich material; however, the quality of compost depends upon the materials used and the conditions maintained at different stages.[43] Several biomass wastes may be utilized as the source of organic matter, including crop residue (stubbles of the crop), waste straw, vegetable refuse, sugarcane trash, and animal waste. Compost generally releases nutrients slowly and stimulates the soil microbial activity, which enhances crop growth. Moreover,[44] reported compost as a pathogenic suppresser, offensive odor reducer, and weedicide.

Vermicompost is one of the types of organic compost manure significantly utilized to enhance the available elements (N, P, K, Ca, P, and Mg), and nutrients improve soil fertility.[29] It is a mixture of worm casts rich in micronutrients, macronutrients, enzymes, and growth-regulating substances, produced by earthworms from different organic residues. The existence of earthworms in the soil endorses the physical conditions of the soil, including aeration, significantly enhancing microbial diversity and plant growth.[45] It transforms the organic wastes into an eco-friendly, valuable product for agriculture, and gardening, and also the remediation polluted areas by preventing soil erosion, retaining soil moisture, improving water infiltration, enhancing water holding capacity, stimulating microbial activity, and most importantly, improving soil fertility.[46]

To maximize the benefits of composting for soil health rejuvenation, it is crucial to follow proper composting practices, including the right mix of materials, proper aeration, moisture management, and temperature control. The composting process should also be completed before applying compost to the soil to avoid potential issues with nutrient availability or the presence of harmful pathogens. Regular application of compost over time can significantly improve soil health and contribute to sustainable agricultural practices. Table 2 represents the effects of compost on physicochemical properties and fertility of the soil.

Rejuvenating soil health using biochar

Biochar, a common porous carbonaceous highly heterogeneous material, is derived from a wide variety of biomasses via pyrolysis.[57,58] Biochar is considered to be a fine-grained and porous substance with a high carbon content that can be tailored to suit all soil types to reach maximum benefit.[59] Biochar amendment soil can significantly alter soil physical properties, including

Table 1. Effects of manure on physicochemical properties and fertility of the soil

Types of manure	Modification (M) if any	Rate of application	Influencing chemical properties	Influencing physical properties	Soil type	Study area	Application	Reference
Animal manure	NA	Applied for 60 days	Increased soil N, P, K, pH, and organic matter	NA	Sandy	Beijing, China	Growing alfalfa (*Medicago sativa* L.)., alfalfa silage in a greenhouse	Huang et al.[25]
Dairy manure	NA	35 t ha^{-1} for 3 years	Increased soil pH, total organic carbon, and total P	Increased water-holding capacity	Portneuf silt loam	Kimberly, Idaho (United States) (42º330 N; 114_21′ W)	NA	Araújo et al.[32]
Swine manure	Biochar	0-5% of the soil	Increased soil N, P, pH, and organic matter	Increased water-holding capacity	NA	Chongzhou, China	Nutrient enrichment in soil.	He et al.[33]
Animal manure	Lignite amendment	30 and 60 t ha^{-1}.	Increased soil organic matter and N	–	Clay loam	Victoria, Australia	Enhanced soil health and improved plant yield	IMPRAIM et al.[34]
Bovine manure [sewage sludge (20%) and green waste (80%)]	Sulfamethoxazole	Applied for 28 days	Increased exchangeable and total organic carbon	Enhanced water-holding capacity	Silt loam (15% clay, 79% silt and 6% sand)	Feucherolles (Yvelines, France)	Enhanced soil health for better crop production	Vieublé Gonod et al.[35]
Poultry and sheep manure	Limestone	1200 mg kg^{-1}	Increased organic matter	Enhanced moisture content and water-holding capacity	Yellow-red soil, yellow-brown soil, sandy loam, and chestnut soil	Beijing, China	Immobilization and phytoavailability of antimony in the wheat	Pervaiz et al.[36]
Solid dairy manure	NA	18 tons ha^{-1} for 14 years	Increased organic carbon, total soil P, and labile soil P contents	Increased moisture content	Portneuf silt loam	Kimberly, Idaho, USA (lat. 42° 33′ N, long. 114° 21′ W)	Improved crop yield of wheat (*Triticum aestivum*), potato (*Solanum tuberosum*), barley (*Hordeum vulgare*), and sugar beet (*Beta vulagaris*)	Pan et al.[37]
Sheep and poultry manure	NA	60 days	Increased TOC and pH	NA	NA	Chang Xing (119°55′34″N, 30°54′27″E) in Zhejiang province, China	Enhanced the phosphorus saturation of soil rice productivity	Khan et al.[38]
Chicken manure and	Wheat straw biochar	2.5% (w/w) about 39 t ha^{-1} 60 days	Increased organic carbon and pH	Enhanced water-holding capacity	Loamy sand	Shanxi province of northern China	Enhanced the bioavailability of heavy metals	Yang et al.[39]

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Table 1. *Continued*

Types of manure	Modification (M) if any	Rate of application	Influencing chemical properties	Influencing physical properties	Soil type	Study area	Application	Reference
livestock (pig) manure	Pig slurry-borne	~5% of the bulk soil massv for 11 years	Increased TOC and soil nutrient	NA	Clayey soil	Victoria, Australia	Redistribution of Zn	Formentini et al.[40]
Pig manure	sepiolite	0- 12% of the soil	Enhanced available nutrient	Enhanced moisture content to about 65%	–	Hechuan Distract in Chong Qing, China	Improved seed germination index increased (15.9 ± 0.5%) and reduced the bioavailability of Cu and Zn	Zheng et al.[41]
Swine manure	Urea, calcium-magnesium phosphate, and potassium chloride	5 years	Enhanced soil pH, organic matter, and nutrient (N.P and K)	Enhanced water holding capacity	Sandy loam	Sixi Town, Jiangxi Province, China (115°07′13″E, 28°15′2″N).	Enhanced phosphorus sorption and rice (*Oryza sativa*) yield.	Li et al.[42]

water holding capacity, pore size distribution and density, soil aeration, and structure resulting in enhancing plant growth and soil workability.[60–62] Carter et al.[46] reported biochar application can reduce soil bulk density and soil moisture content and increase soil water holding capacity, which significantly influences most of the plant's growth. Adekiya et al.[63] conducted a two-year field experiment using biochar (from hardwood) amendment on a sandy loam soil (Alfisol at Owo, southwest Nigeria), resulting in significant improvement in soil physical properties (including porosity, bulk density, dispersion ratio, soil aggregates, infiltration rate, and moisture content) and chemical characteristics (including soil pH, soil organic matter, cation exchange capacity and nutrients such as N, P, K, Ca, and Mg) resulting into a high yield of cocoyam (*Xanthosoma sagittifolium*).

Studies by Adekiya et al.[63] revealed that biochar modification significantly influences the nutrient status of the soil along with the overall soil health improvement resulting in enhanced crop yield. However, many of the published literature reported high variations of the factors, including biochar dose, biochar characteristics, initial soil properties, and working environmental conditions. Significantly positive crop and biomass yields were reported for the biochar produced from hardwood, crop residue, poultry litter, and paper pulp. Neemisha and Rani[29] reviewed literature from 57 field experiments and 59 pot experiments from 21 countries worldwide and found a ~11% average increment in crop productivity. The highest crop yield was documented for legume crops (30%), followed by vegetables (29%), grasses (14%), wheat (11%), corn (8%), and rice (7%). A meta-analysis by Singh et al.[30] showed significant enhancement of aboveground productivity, crop yield, and soil microbial diversity along with an increase in soil N, P, K, and available organic carbon. However, some of the studies also raised concerns about the heavy metal contamination of biochar (derived from sewage sludge and spent biochar), which might contain toxic metals that limit the land application due to the chance. Table 3 represents the effects of biochar on physicochemical properties and fertility of the soil.

METHODS OF BIOCHAR PRODUCTION

Studies on biochar have been on the rise since the first article on biochar was published in 2000 by Karaosmanoğlu et al.[76] The search for articles on biochar was analyzed retrieving records from titles and abstracts on such studies. A total number of 40,857 articles have been found for biochar (from 2000 to 2024), among them 1874 articles on soil fertility enhancement using biochar and only 221 articles on the enhancement of soil fertility and microbial diversity using biochar (Figure 1). The statistics in Figure 1 represented the number of biochar-based articles, enhancement of soil fertility, and enhancement of soil fertility and microbial diversity published in different articles from 2010 to 2024 (data up to 26th August 2024). Figure 1 indicates a significant increment of biochar-based articles and biochar-based articles for the enhancement of soil fertility and has been increasing globally in the first and second decade of the 21st century.

Several methods can be used to produce biochar namely; pyrolysis, gasification, torrefaction, and hydrothermal carbonization (HTC). However, this section is fully focused on pyrolysis

Table 2. Effects of compost on physicochemical properties and fertility of the soil

Composting feedstocks (CF)	Modification (M) if any	Rate of application	Influencing chemical properties	Influencing physical properties	Plant/seed species	Study area	Effects on plant growth and yield	References
Rice straw	Rice straw derived biochar	1:2 (CF:M)	Increased soil N, P, K, CEC, organic matter, and exchangeable Ca^{2+}	–	Canola	El-Gharbia Governorate, Egyp	Increased plant growth and the dry weight	Mahmoud et al.[47]
Phragmites australis	*Phragmites australis* derived biochar	1.5% (w/w) of the CF	Increased soil organic carbon but decreased soil ammonium nitrogen content	Increased water-holding capacity	*Phragmites australis*	Shandong Province, China	Increased photosynthesis rate (51.4%) and plant biomass (35.5%)	Liang et al.[48]
Poultry manure	Bamboo derived biochar	5–20% (w/w) of the CF	Increased soil N, P, K, CEC, and organic matter	Increased pile porosity and air permeability.	NA	Beijing, China	Increased plant productivity	Liu et al.[49]
Cow dung + hydrilla + Sawdust	Wood derived biochar	5% (w/w) of the CF	Increasing pH and total N by 45%.	Decreased air-filled porosity by 39%.	–	Guwahati, Assam, India	Increased plant growth	Jain et al.[50]
Rice straw	Agricultural waste derived biochar	10% (w/w) of the CF	Increased soil TOC, pH, EC, organic matter, N, P and K	Increased water-holding capacity	NA	Changde City, Hunan Province, China.	Increased plant growth and productivity	Tang et l.[51]
Sewage sludge + Wheat straw	Wheat straw derived biochar	8–12% (w/w) of the CF	NA	NA	NA	Yangling, Shaanxi Province, China	NA	Awasthi et al.[52]
Pig manure	Maize straw derived biochar	2% (w/w) of the CF	Increased soil pH, DOC CEC, TOC, and	Increased soil permeability	carrot (*Daucus carota*)	Nanjing, China	Increased dry weight	Xiang et al.[53]
Plant straw and chicken manure	Palm fruit bunches	48 t ha^{-1}	Increased soil nutrient availability CEC, C, N, and WHC	Increased water-holding capacity	Cucumber	Yinchuan, China	Increased root length, plant biomass, and fruit yield	Cao et al.[54]
Farm manure	Garden peat	2% (w/w) of the CF	Increased soil organic matter, EC, total N and K	NA	Wheat	Multan, Pakistan	Increased grain weight and grain yield	Qayyum et al.[55]
Sewage sludge, Food and green waste, algae waste, livestock waste,	NA	NA	Increased soil pH, organic matter and nutrient content	NA	NA	Santiago de Compostela, Spain	NA	López et al.[56]

Table 3. Effects of biochar on physicochemical properties and fertility of the soil

Biochar feedstock	Biochar Modification	Rate of application	Soil Type	Influencing chemical properties	Influencing physical properties	Study area	Application	References
Different biomass	NA	Eight-years (2014–2021)	–	Increased pH, CEC, and SOC	Increased bulk density and Soil porosity	Liuyang, Hunan Province, China (28°19′ N, 113°49′ E).	Improve maize productivity and mitigate nutrient loss	Yan et al.[64]
Agricultural residue	Organic fertilizer	1 year (2019)	Sandy loam, silty loam, loam, silty loam, silty, and silty clay loam (6 sites)	Increased CEC, Fe, Al, pH (0.1–1.1↑), CEC (10.8–30.3↑), and SOC (11.8 g/kg ↑)	NA	Zhejiang Province, China	Reduced soil water-dispersible colloidal phosphorus	Jin et al.[65]
Cornstalk Orange peel Peanut hull	H_3PO_4 and KOH modification	25 g biochar kg^{-1} soil	Silty loam	Increased CEC and SOC	Improved water holding capacity and reduced soil bulk density	Bahman Blvd., Tabriz, Iran	Soil water retention	Ghassemi-Golezani et al.[66]
Sheep manure and biogas slurry	Biochar-blended with organic fertilizer	Experimental sites were set up in March 2018, and samples were collected in November 2019	Alkaline soil	Increased pH, organic matter, Fe, Al, and Ca	NA	Tian Tai, Lu Qiao, Zhou Shan, Zhu Ji, Ma Zhan, and Tong Xiang, Zhejiang Province, China	Improvement of phosphorus availability and increasing rice yield.	Jin et al.[67]
Corn stalk	Fe and Zn compounds modification	42 days	Acidic and alkaline soil	Increased soil pH, CEC, and dissolved organic carbon	Increased soil porosity	Xinxiang City, Henan Province, China	The release of alkaline salts from the biochar into the soil	Yang et al.[68]
Rice straw	Iron oxide	30 days	Alluvial soil	Increased soil pH soil nutrients (such as Ca, K, Mg, Na, and P)	NA	Deganga block, North 24 Pargana (22°41′13.2″*N* 88°38′ 40.6″E), West Bengal, India	Minimize arsenic bioavailability and nutritional enrichment	Majumdar et al.[69]
Corn straw	NA	30 days	Black soil	Increased soil pH and organic carbon	NA	Acheng, Harbin, Heilongjiang Province, China	Phosphorus adsorption performance and storage capacity of black soil	Xue et al.[70]
Wheat straw	NA	Two years	–	Increased soil pH, ammonium nitrogen (NH_4^+-N) and nitrate nitrogen (NO_3^--N) content	NA	Shaanxi Province, China	Soil nutrients retention and rice growth	Han et al.[71]
Wheat straw	Sulfoaluminate cement	0-5% (w/w)	sand 15.46%, silt 83.33%, and clay 1.21%	Increased soil pH, EC, and organic matter	NA	Shaanxi Province, China	Inhibition of Pb-Cd mobility in the soil-romaine lettuce	Han et al.[71]
Corn stalk	nZVI/nFe_3O_4	NA	NA	Increased soil pH	NA	Chengdu, China	Increased salinity and Immobilization of Pb in contaminated soil	Li et al.[72]

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Table 3. *Continued*

Biochar feedstock	Biochar Modification	Rate of application	Soil Type	Influencing chemical properties	Influencing physical properties	Study area	Application	References
Rice straw	$FeCl_2$ and $FeCl_3$ modification	3 years	Saline-alkaline soil	Increased soil pH and CEC	NA	Li Wang Village, Dongying, Shandong, China	Soil fertility	Wu et al.[73]
Wood chips	Compost and Manure	0-10% biochar (w/w)	Loamy sand	Increased soil pH and organic matter	Decreased bulk density and increased total porosity	Zvěřínek, Prague, Czech Republic (50°149′N, 15°026′E)	Improved soil water retention. reducing nutrient leaching and increase crop yield.	Seyedsadr et al.[74]
Agricultural waste (Straw-derived biochar)	Manure	0.15 ± 0.002 mg kg^{-1}	Loamy soil	Increased soil nutrient content	Increased soil moisture	Nanjing City, China	Reduce the ecological risk of PBDE and promote nitrogen cycling by removing 2,2′,4,4′-Tetrabromodiphenyl Ether	Xiang et al.[75]

and gasification.[77–80] Table 4 presents a summary of various methods of biochar production and the conditions for which they were conducted.

Pyrolysis

Pyrolysis is a thermochemical approach that uses extremely high temperatures and in the absence of oxygen and under atmospheric pressure to produce solid in the form of biochar, liquid in the form of bio-oil, and gas in the form of synthesis gas (syngas).[99] Several types of pyrolysis exist namely; slow pyrolysis and fast pyrolysis. The slow pyrolysis process can be achieved by putting feedstock into a paint can filled with a nitrogen purge at 1 L/min and a thermocouple for temperature measurement.[99,100] The fast pyrolysis is performed at a much larger flow rate of 5 kg/h than that of slow pyrolysis while a constant temperature of 500°C is maintained in both cases.[81,100] Other types of pyrolysis that exist include flash pyrolysis and catalytic pyrolysis which depend upon various operating conditions. Hodgson et al. carried out the Taguchi optimization of pyrolysis process conditions to maximize char production and heavy metal removal.[101] They found out that chars produced from Loium Perenne gave the greatest adsorption among other biochars removing over 90% of heavy metal from wastewater. Lee et al.[102] achieved fast pyrolysis at a temperature of 450°C and successfully correlated the impact of the resultant biochar on the soil amendment by using the oxygen-to-carbon ratio as an increase was tantamount to improved soil amendment and CO_2. The pyrolysis method has the following advantages namely; it can be operated in the open air without the aid of diluents such as nitrogen or other inert materials, it is exothermic, and has a higher throughput as feedstock can be added continuously.[84] Another type of pyrolysis is microwave-assisted pyrolysis (MAP) which is an effective technology for the production of biochar but it is grossly limited when scaled up for industrial purposes.[103] This has been addressed by Selvam and Paramasivan[103] as they optimized the various factors ranging from feedstock amount to exposure time to know the most influential for biochar production. They concluded that the most influential of all were feedstock size, microwave exposure time, feedstock quantity, and microwave capacity.

Gasification

Gasification is a promising thermochemical process where a solid carbonaceous material is converted into a flammable gaseous fuel called syngas. The syngas can be used as fuel in an internal combustion engine, generating both thermal and electrical energy. Micro-scale gasification plants, in particular, can provide distributed energy that matches local energy demand and feedstock production.[104] This is the process of biochar production from biomass produced at the rate of 3 kgh^{-1} in a fluidized bed reactor taking place either under air-blown or steam/oxygen-blown conditions.[81] The gasification process can affect the biochar produced by increasing the pore surface areas of the biochar. This effect has been reported for both raw and acid-treated biochars.[105] Another aspect of gasification is Steam gasification which is a process where steam is used as a reactant to convert a substance into a gas. According to Chen et al.,[87] steam gasification is used to convert bio-slurry into

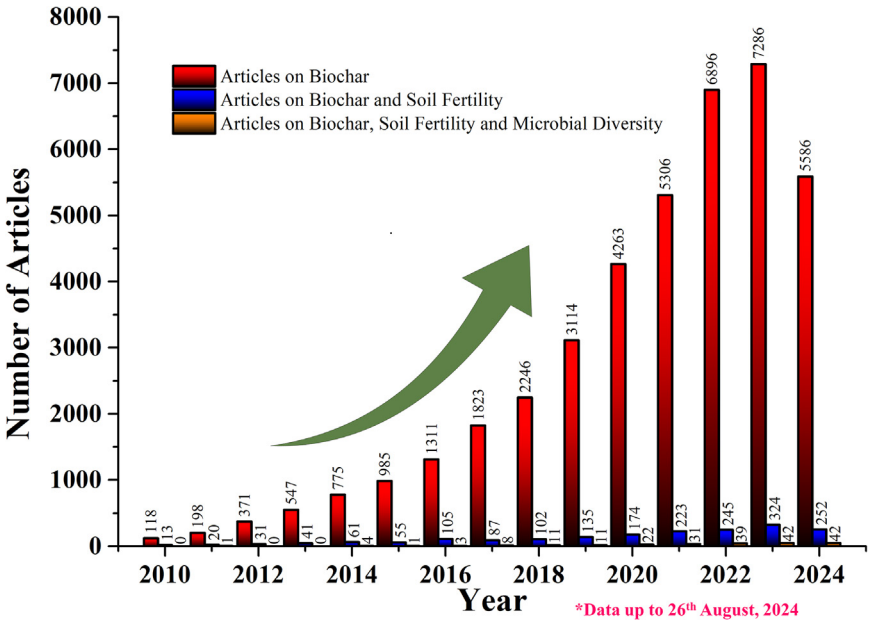


Figure 1. Temporal trends of biochar-based articles published from 2010 to 2024

hydrogen-rich syngas. The process involves reactions such as char-steam reactions and steam reforming reactions, which are accelerated by catalysts such as $La_{0.8}Ce_{0.2}FeO_3$. Moreover, during steam gasification, the highly heterogeneous and more reactive disordered carbon structure is used up first leading to the enrichment of the less reactive ones, however, in the presence of catalysts, the reverse becomes the case as gasification will focus more on the localized catalyst sites thus altering the reaction mechanism. So, it can be concluded that gasification can be affected by several factors namely; catalyst amount, carbon structure, and then the interaction of these two properties. In the work of Allesina et al.[104] cotton crop residue was processed and converted into briquettes, which were then used as fuel in a micro-scale gasifier, an All Power Labs PP20. This gasifier transforms solid wood biomass waste into electrical and thermal power. For every kg of dry wood biomass, the gasifier generates 0.8–0.9 kWh of electrical energy, along with 50 g of biochar. This approach generated over 4 MWh and about 130 kg of biochar per year from a 1-ha field. The produced biochar can be used as a soil amendment, reducing the need for fertilizers by up to 50%. This model promotes sustainable development by providing clean, affordable energy and improving soil conditions. It has also been confirmed that biochars produced from wood waste via gasification are highly efficient in the removal of dyes from wastewater.[106]

Torrefaction

This is an attractive thermochemical transformation approach for the conversion of biomass to a uniform and hydrophobic fuel which has better heating value, increased combustion characteristics, and improved grindability.[93] This method of biochar production upgrades biomass of microalgae origin to meet the coal properties by increasing the calorific value of the produced biochar.[107] During the torrefaction of forest residues, it was observed that the decomposition of the initial biomass to produce an intermediate solid mass had a higher conversion rate than the subsequent degradation of the intermediate.[108] This process converts raw biomass into biochar at a temperature range of 200°C–300°C at atmospheric pressure and inert conditions thus producing hydrophobic species.[107] The wet approach, however, takes place at a much lower temperature but produces a high-energy biochar. Hybrid materials are used to produce biochar via torrefaction, for instance, Hanoglu et al.[94] combined textile fiber (blends) to produce biochar at a temperature range of 300–400°C. One of the major milestones in this work is the fact that 100% of the polyester fiber cannot produce biochar due to its high thermal stability.[94] These biochars are different from those of other sources in that they are low in both ash and sulfur contents after torrefaction. It is worthy of note that the process of torrefaction leads to low throughput which results from the fact that the process proceeds at a relatively low temperature compared to other methods. To improve the throughput, it is now necessary to add microwave heating to the conventional heating method for the process.[109] The features of biochars produced from microalgae, forest residue, and hybrid materials via the torrefaction process have been summarized in Figure 2.

Hydrothermal carbonization

This is a process for producing high-performing biochars by dehydration and decarboxylation which leads to the complete decomposition of the hemicellulose and the partial decomposition of the cellulose in addition to the oxygen functional group in the product.[91] This is gaining interest as one of the many approaches for the production of biochars. This process involves the combination of moderate temperature (250–350°C) and high pressure (3–11 MPa) in an oxygen-free environment using subcritical water as a medium.[110] Kim et al.[92] showed that the optimum temperature for this process must be around 210 to 240°C for the reason the hydro chars produced at this temperature are of high calorific value (26–27 MJ/kg).[92] One of the key advantages of this operation is that it can save energy whilst producing biochars from wet biomass efficiently and ultimately minimizes the cost required. The biochars produced from the operation are better referred to as hydrochars.[80] A diagram that gives a detailed description of what this process consists of is shown in Figure 3. It has been shown that hydro chars (from HTC) are better in quality than their counter parts pyrochars (from pyrolysis). Hydrochars have higher heating value, oxygen functional, carbon content, and volatile matter. However, it does have a lower surface area but still has a higher adsorption capacity than most pyrochars, for instance, for the adsorption of copper

Table 4. Summary of biochar production processes, sources, and operating conditions

Source	Process	Peak Temp. (°C)	Advantages	References
Corn stover	Fluidized Bed Fast Pyrolysis	500	The reaction atmosphere is oxygen-dependent	Brewer et al.[81]
Pine Sawdust	Pyrolysis/gasification	600	Higher CO_2 reduction efficiency (CRE) and feasible biochar briquettes, reduction of tar yield	Liu et al.[82]
Pine Sawdust, Maize Straw, Surgar cane bagasse	Pyrolysis	200–700	The aromaticity of biochar enhanced at above 400°C	Luo et al.[83]
Corn stover, wheat straw and wheat straw treated with glycerin	Pyrolysis	400–700	Normal atmospheric conditions are prevalent, more amenable to scale up and peat moss replacement in horticultural application	Peterson et al.[84]
Hickory wood, bagasse, and bamboo (BB)	Pyrolysis	200–600	Increased carbon content, thermal stability and lower pH	Sun et al.[85]
Cherry Pomace	Pyrolysis and gasification	300–1000	Conversion of wet residues, production of a valuable intermediate products for final production	Wądrzyk et al.[86]
Boslurry	Steam gasification	800	Maximum hydrogen yield under optimum conditions	Chen et al.[87]
Wood chippings	Gasification	200–500	Carbon storage in stable form and increased recycling rate	Sirico et al.[88]
Coconut shell	Gasification	50–800	Improved pore structure and hence promising industrial and agricultural field application	Wang et al.[89]
Rice husk	Hydrothermal Carbonization	180	Improvement of surface area and porosity for adsorption	Hossain et al.[90]
Walnut shell	Hydrothermal Carbonization	240	Increased aromaticity, pore content and surface area.	Kang et al.[91]
Exhausted coffee residue	Hydrothermal Carbonization	180–330	Increased calorific value, carbon content and fixed carbon	Kim et al.[92]
Forest residue	Torrefaction	220–300	Increased carbon content, reduced hydrogen and oxygen contents	Bach et al.[93]
Textile fibers	Torrefaction	300–400	Energy-intensive solid fuel with a negligible amount of ash and sulfur	Hanoğlu et al.[94]
Elephant grass	Retort carbonization	371	Improved carbon content and specific surface area	Adeniyi et al.[95]
Almond leaves	Retort carbonization	494	Globular biochar surfaces with smaller interstitial voids	Ighalo et al.[96]
Mesquite (*Prosopis juliflora*)	Retort carbonization	500	Increase in fixed carbon, ash content, HHV, surface area, and pore volume.	Chandrasekaran et al.[97]
Sponge gourd pod	Retort carbonization	100	Porous biochar as confirmed by SEM and BET analysis	Adeniyi et al.[98]

ions and iodine, it has over 100% greater impact than pyrochars.[50] The reason for this superior performance can be related to the fact that after the thermal treatment at a much lower temperature, the various hydrocarbons will remain intact. Hossain et al.[90] achieved a biochar yield of about 57.9% using the HTC approach with temperature, pressure and time kept at 180°C, 70 bars, and 20 min, respectively. Other properties such as bulk density, heating values, and thermal profile were all favored for biofuel application requirements. Improved biofuel can be obtained by the targeted carbonization of biomass, for instance,

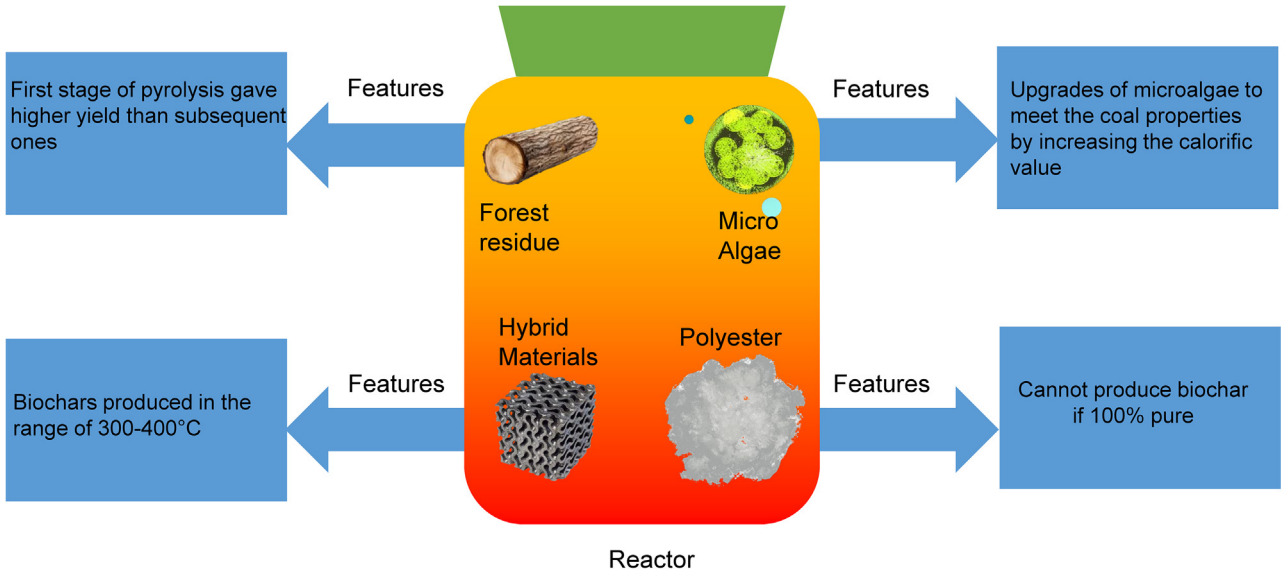


Figure 2. The features of biochars produced from microalgae, forest residue, hybrid materials, and polyester materials via torrefaction

Kim et al.[111] focused their work on the targeted carbonization of cellulose, xylan, and lignin via HTC between 150°C and 280°C.

Retort carbonization

Retort carbonization is a special carbonization technique where the flue gas from the biomass conversion is re-combusted in the heating zone to help the system achieve a higher process temperature at a lower energy input.[112] The retort carbonization technology is interesting as it does not require any electrical heating (though this can be implemented). The heating process (when not electrically powered) is self-regulating.[113] The technology was first publicized by Adam[114] in the development of the ''Adam retort.'' Many recent studies have successfully employed the technology for the production of biochar from biomass waste.[115] employed retort carbonization to convert neem leaves to biochar. They observed that the process could yield a product with a high specific surface area when operated in semi-batch mode as opposed to the conventional batch mode in which it is usually operated (Figure 4).[97] examined *Prosopis Juliflora* conversion by heating in a natural updraft retort kiln to a temperature of 500°C. Their study revealed a 29.6 wt % biochar yield and about 50% retention of the energy content in the original biomass. Ighalo et al.[116] utilized this process for the production of biochar from *Delonix regia* pods. At a relatively low process temperature of 375°C, a 29.5 wt % biochar yield was achieved, and the product had high surface roughness.

FACTORS INFLUENCING BIOCHAR PRODUCTION

Controlled operating conditions are indispensable for getting real fantastic results in both pyrolysis and gasification both for soil amendment and CO_2 sequestration. In the production of

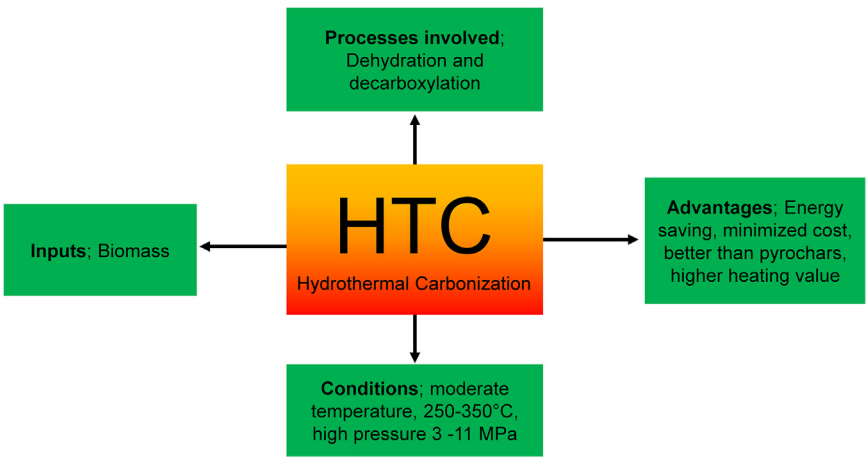


Figure 3. Description of a hydrothermal carbonization (HTC) system

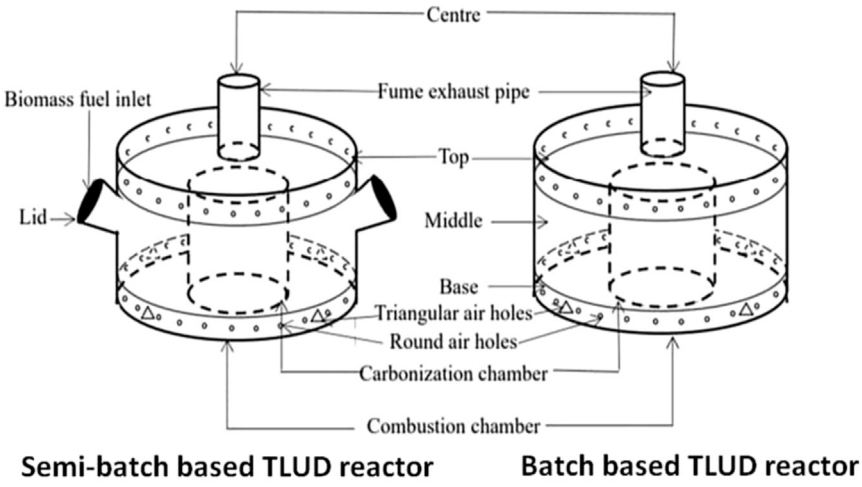


Figure 4. Description of semi-batch and batch retort carbonization systems

biochars depending on the goal to be realized by researchers operating conditions could be controlled to achieve the goal. The work of Pelaez-Samaniego et al.[117] which covered the gasification of wood species suggests that for gasification operations, it is possible to control the factors affecting the optimization of the process for the simultaneous production of producer gas and char and biogas. In the work of Liu et al., the object was to increase the carbon reduction efficiency (CRE) so the CO_2 to biomass ratio was 135 g/kg by experimenting with 10% CO_2 and at 600°C.[82] Also, the work of Luo et al. demonstrated that the production temperature and atmosphere synergistically influence the properties of biochars and also increase their yield. However, a decrement in biochar yield and volatile matters was recorded at lower temperatures.[83]

Effect of feedstock type

The feedstock has its natural effect on the biochar production process. The proximate and ultimate composition of the biomass plays a role in the hydrophilicity, aromaticity, and mineral content of the biochar. It also affects the porous structure of the biochar. Some research has shown that the feed can be used concerning the other additional components which may include steam in which case the charge into the reactor will be defined as a ratio of the steam amount to the carbon amount in the biomass. For instance, in the work of Chen et al.,[87] the feedstock was measured as a ratio of the steam to carbon amount present in the bioslurry which was referred to as water to carbon molar ratio (WCMR). At a given temperature of 800°C, the yield of hydrogen and carbon conversion followed a similar trend of increase and decrease with each attaining their maximum values at a WCMR value of 2. The increase was possible since the higher WCMR favored the steam reforming reaction and the water gas shift reaction (WGSR), respectively.

Effect of temperature

Temperature is one of the most fundamental factors in the pyrolysis of biomass. It presents two possible extremes for the operation which are; the 200°C mark which is the commencement of the actual pyrolysis process and the 1000°C mark which is the point at which the biomass is completely degraded.[99] Determination of the optimal parameters via multi-parameter decision analysis for the process of biochar production is indispensable. In their research, Evaristo et al. found out that biochar produced from spent coffee ground (SCG) and brewers' spent grains (BGS) were obtained by decision analysis made on the temperature which was fixed between 700°C and 300°C and this had a great influence on the chemical properties of the biochars more than on the physical properties.[118] Sometimes temperature will not be counted as a controlled variable this is especially true for processes that are self-sustaining and after the initial ignition of the feedstock.[84] It should however be noted that temperature is a dynamic property and hence depends on the feedstock type and the position of the pyrolysis front. Graduation in temperature causes biochar materials to change their behaviors such as increasing aromatization with increased temperature as was detailed in the study of cherry pomace by Wadrzyk et al..[86] For temperature effect on biochar production Chen et al.[87] highlighted three major factors that determine the amount of gas to be obtained namely; the increase in heating rate as temperature increases which leads to effective thermal decomposition, second, higher temperatures improve all endothermic processes leading to higher hydrogen yield and third, temperature enhances CO yield as it reverses the WGSR.

Effect of heating rate

The heating rate is the rate at which the temperature of the biomass conversion is ramped from ambient to process temperature. Studies have shown that the heating rate can significantly influence the quality of the biochar produced. For example, in pyrolysis, the heating rate is used to classify the process and determine if its slow pyrolysis, fast pyrolysis, or flash pyrolysis.[112] When the heating rate is high, the process tends to evolve more fluid phase products (gases and liquids), and solid phase products are favored. This has been observed experimentally by[119] for sugarcane bagasse pyrolysis. The reduced solid product at a fast heating rate is due to the rapid depolymerization of biomass.[120] However, when slower heating rates are employed, the residence time of the feedstock in the process could be longer and energy consumption can become higher.[121]

BIOCHAR CHARACTERIZATION

Depending on the raw materials and pyrolysis procedures, biochar has outstanding physicochemical qualities, including its high density of negatively charged ions, high pH, high carbon content, high surface area, high porosity, recalcitrant carbon content, and big cation adsorption ability.[122,123] Biochar lessens

the toxicity of pesticides to soil organisms, hence enhancing soil fertility and agricultural yields.[124] Biochar can be used as a soil amendment to improve degraded soils, increase water use efficiency, increase nutrient availability, remediate and/or protect from organic or inorganic pollutants, and reduce greenhouse gas emissions (nitrous oxide, methane).[125–127] Growing soil organic carbon and lowering atmospheric CO_2 levels have both been achieved by incorporating biochar into the soil. A biochar supplement can increase the net soil surface area and change bulk density.[128] Among its notable qualities are its large surface area with several functional groups, nutrient-rich composition, and ability to act as a slow-release fertilizer.[129] Previous research has demonstrated that adding biochar to soils has a variety of effects due to its increased surface area, including better soil microbial adhesion.[130] Results revealed that more than one type of biochar had varied effects on soil characteristics, microbial community structure, and diversity.[75] Therefore, one form of biochar may not be enough to improve vegetated soil over the long term to keep it in the best possible condition for plants.[131] The amount of positively charged binary heavy metal ions that can pair with anionic functional groups on the surface of bacteria increases as H^+ ions in solution are reduced, enhancing the capacity of bacterial cell walls to bind heavy metals.[132] The characteristics of biochar for soil fertility and microbial enhancement are thus described in this section.

Chemical composition

Biochar yield is significantly influenced by the moisture content of the biomass. Biomass is unsuitable for the formation of biochar if it has a moisture content of more than 30% leading to increased energy and time requirements to achieve the necessary pyrolysis temperature.[133] According to research, biomass with 5–20% moisture is the optimal moisture range because biochar output declines as biomass moisture content increases; thus, feedstock with less than 10% moisture content has been suggested for a successful pyrolysis procedure.[134] To ensure long-term storage stability and prevent mold formation, biochar should ideally have a low moisture content.[121] Furthermore, the high temperatures and moisture content of paddy fields facilitate the decomposition of straw.[135] The treatment of rice straw at various moisture contents (80%, 60%, and 40%) resulted in significant reductions in Cu and Pb concentrations in the roots and shoots of maize plants, particularly under the highest moisture content (80%) when rice and rapeseed biochar was added at 5% application rates.[136] In Table 5, palm kernel, empty fruit bunches, mesocarp fibers, and wheat straw gave moisture content ranging between 0.12 and 7.9%.[137,139] Light torrefaction conditions resulted in a yield of 70–78%. However, when exposed to the torrefaction process, a mass loss of 45–55% of these biomasses was noted.[137] A huge moisture content was observed in apple, pear, and plum wood chips ranging from approximately 41-45%. The best biochar production efficiency was found at a final temperature of 400°C and a heating rate of 15°C/min. Conversely, rapid pyrolysis at high ultimate temperatures (600°C and 100°C/min) was employed to produce the biochar with the maximum carbon content and heating value with an additional 10% loss.[142] The findings from poultry litter and sugarcane straw at 60% demonstrated that raising the temperature of pyrolysis promoted the stability of biochar independent of feedstock; yet, the effects of temperature and feedstock on agricultural properties differed greatly.[140] Biochar made from animal dung has greater potential as a source of nutrients than as a method of sequestering carbon. The increased concentration of alkaline metals (Na, Mg, K, and Ca) in soil heating rate biochars (at pyrolysis temperatures of 450°C) suggested their use as soil additives for improving soil fertility and decreasing soil acidity, even at extremely low moisture contents of 0.6–0.9%.[144]

When compared to the control treatment, soil respiration increased because of the higher proportion of volatile matter found in the biochar made from poultry litter. This indicates a larger amount of readily degradable carbon source that the microorganisms can use.[140] Empty fruit bunches, mesocarp, and palm kernel shells showed volatile matter of approximately 70–72%,[139] poultry litter, sugarcane, rice hull, sawdust showed approximately 25–60%,[140] and palm kernel shell gave 11.7%[145] (Table 5).

Fixed carbon refers to the recalcitrant carbon that remains in biochar composition following thermal degradation brought on by pyrolysis.[146] The amount of fixed carbon is more influenced by the type of feedstock used than by the pyrolysis temperature during production. The higher the amount of ash in the feedstock, the less impact increasing fixed carbon at higher temperatures has.[140] A high range of fixed carbon (73–88%) was observed from wheat straw, timothy grass and pine wood, palm kernel (Table 5).[144,145] Contrastingly, wheat straw showed 13–20% in another study.[143]

Crop leftovers have more ash than biochar made from wood due to the higher nutrient concentration in the crop feedstock, Despite being rare, sugarcane straw biochar has ash levels that range from 11 to 13% which increases proportionally with the temperature rise. The surprising decrease in ash concentration in sugarcane straw could be attributed to the element P, K, and S volatilization, which can happen at temperatures as low as 500°C.[140] The solid biomass-derived biochar, which is made from materials including nutshells, wood, bamboo, and seeds, likewise has a low ash content and a high porosity and surface area. Other biomasses with a medium ash content were discovered, including bark, agricultural wastes, and high-quality green waste under the condition that they had a low level of contamination from soil, plastics, and metals. In contrast, sludges, manures, wastepaper, municipal waste, and biochar made from rice husks have higher ash levels and consequently higher phosphorus content than other varieties.[147–149] Additionally, the pH of the soil may rise due to the liming effects of biochar and high ash contents.[150] The larger ash content produced can be due to the higher pH of the rice husk biochar.[151]

Inorganics and pH

The pyrolysis temperature is an important factor to take into account while creating biochar because it directly affects some qualities of the material, such as its pore structure, surface characteristics, nutrient makeup, and functional groups.[152,153] Sunlight and sun drying are regarded as environmentally favorable pretreatment methods for pyrolysis.[134] While thermochemical conversion is more appropriate for low-moisture (less than 10%) agro-residues, anaerobic digestion, and hydrothermal

Table 5. Chemical composition of biochar for soil fertility

Residue-derived biochar	Moisture content (%)	Volatile matter (%)	Fixed carbon (%)	Ash (%)	pH	Total metal (%)	Reference
Empty fruit bunches, Mesocarp fiber Palm kernel shell	0.67 0.37 0.12	–	–	0.0167 0.0350 0.0104	–	–	Sabil et al.[137]
Empty fruit bunches, Mesocarp fiber Palm kernel shell	7.16 7.90, 6.47	68.58 68.97 72.04	17.30 14.97 19.61	6.96 8.16 1.88	–	C:44.80, 45.67, 49.5; H: 7.30, 7.29, 6.82; N: 0.65, 2.41, 0.45; S: 0.47, 0.44, 0.37; O: 46.78, 44.19, 42.86; (Na: 96, 45, 26; K: 13396, 4936, 1201; Ca: 1477, 3126, 441; Mg: 1242, 2031, 448; Si: 158, 172, 85; Fe: 371, 471, 209) mg/kg	Chew et al.[138]
Oil palm fiber, palm kernel shell	<20	81.03	14.87	<10	–	C: 53.78; H: 7.20; S: 0.51; O: 36.30	Yahayu et al.[139]
Poultry litter, sugarcane straw, rice hull, sawdust pyrolyzed at 350°C, 450°C, 550°C and 650°C	60	42.1–60.8; 43.8–50.1; 24.2–28.0; 29.1–54.0	–	13.3–24.2, 40.4–42; -	9.43; 8.95; 8.5; 7.4	N: PL: 1.46%; SC: 1.43%; RH: 0.02; SD: 0.3%. Fe: 0.45; 6.49; 0.06; 0.48; Mn: 0.05; 0.09; 0.04; 0.05; Zn: 0.08; 0.03; 0.01; 0.02	Feola Conz et al.[140]
Eucalyptus wood at 450,550, 650, 750, 850, 950	–	11.9	–	0.78	8.3	–	Fernandes et al.[141]
Apple, pear, plum wood chips	44.86, 40.9141.76	–	–	1.96, 1.33, 2.42	–	–	Kazimierski et al.[142]
Wheat straw	8.45	66.8	19.1	5.75	8.91	H: 5.68; O: 60.1; C: 34.22 g/kg	Khan et al.[143]
wheat straw, timothy grass, and pinewood	0.6–0.9	7.2–8.9	86.3–88.4	3.1–4.6	–	N: 0.2–1.9; C: 63.7–81.4; H: 2.3–3.7; O: 15.3–31.5; S: 0.03°C–0.14OC: 669.15 g/kg	Mohanty et al.[144]
Palm kernel shell	–	11.7	73.1	15.2	–	C: 75.3; H: 0.86; N: 0.79; O: 7.9;	Dominguez et al.[145]

carbonization or liquefaction are better suited for high-moisture (up to 80%) agro-residues, such as crop residues, municipal trash, algal biomass, animal, and human wastes, into valuable products on a wide scale without the need for pre-drying.[154–156] However, it has been noted that the hydrothermal carbonization process is energy-intensive and produces potentially harmful residues such as ash and tars.[154] While, biochar made at low pyrolysis temperatures (300°C–400°C) can contain potentially more polluting species such as metals, polycyclic aromatic hydrocarbons (PAHs), and pyrophosphate.[157] Higher temperatures during pyrolysis resulted in an overall rise in carbon content and corresponding decreases in nitrogen, hydrogen, oxygen, and sulfur. The date seeds that were pyrolyzed at 800°C had the maximum carbon concentration, with 97.99% of carbon content.[153] The specific C, P, K, Ca, ash content, pH, and specific surface area of biochar normally increase with pyrolysis temperature whereas the N, H, and O content typically decreases.[158,159] At 400°C, biochar maintains volatile and readily labile chemicals, but biochar produced at 600°C has a high recalcitrant behavior.[160] Biochar surfaces are alkaline because they include negatively charged carboxyl, hydroxyl, and phenolic groups. Soil pH is typically increased through biochar treatment, which can neutralize acidic soil.[161] All pyrolyzed materials exhibit increases in pH, which can be explained by the way temperature affects the release of volatile stuff that contains acid functional groups and concentrates ash contents, raising the pH.[140] The pH of most of the biochars ranges from 7.4 to 9.4 as shown in Table 5.

Surface area, porosity, and morphology

The benefits and functions of biochar as stated in Table 6 are possible due to biochar's surface area, high aromaticity, and pore-size distribution, among other properties.[125,126] Increased surface area, pore size, and process temperature were all related to the evaporation of organic compounds.[145,153] Increasing pyrolysis temperatures were found to cause a decrease in the porosity and surface area of biochar made from rice straw.[30] Contrary to this, the impact of the slow pyrolysis process' temperature range (300°C–750°C) on the surface area of samples of wood biochar was examined. At the highest temperature of the operation (600, 750°C), the maximum surface area of the wood biochar sample was achieved.[190] Activated char samples showed a larger surface area when compared to inactivated char. Likewise, more surface area and porosity are present in soils with biochar amendments than in soils without biochar.[161,191] The application of rice husk biochar can change the physico-chemical characteristics of paddy agriculture soil due to its distinct physical characteristics (high surface area and porous internal structure).[192] The greater nutrient, labile carbon, and water retention capabilities with biochar could be responsible for the increase in microbial diversity.[161] In addition to aiding in nutrient retention by binding significant anions and cations, biochar's greater surface area and pore space make it a habitat for soil bacteria that are good for soil health.[193] Contrary to popular belief, straw-based feedstocks have a high cation exchange capacity, animal dung feedstocks produce biochar with the largest N and P content, and wood-based feedstocks are thought to produce biochar with a large surface area.[158] The surface area of some biochars have been reported such as for palm kernel and rice husk (184.29 and 1.69 m^2/g),[151] maize straw (6.82 m^2/g),[167] corncob (8.57 m^2/g), rice straw (37.5 m^2/g), rice husk (209 m^2/g), cow manure (40.2 cm^2/g),[168] chitosan straw and kitchen waste (1.36 and 71.45 m^2/g),[171] rice straw (13.31 m^2/g),[157] wheat straw gasification, fresh straw (75 m^2/g),[186] manure (0.75 m^2/g),[187] wheat straw and timothy grass (166–203 m^2/g),[144] and palm kernel (1842 m^2/g).[189] The effect of pyrolysis temperature was not significant on the surface areas of the various biochars as shown in Table 6.

The abundance and composition of soil microorganisms, soil microbiology, and straw decomposition all directly affect the release of nutrients, grain yield, and rice quality. These factors also indirectly affect morphology, root growth, and activity. As a result, the rice quality and grain yield are affected by the build-up and transportation of rice's dry matter.[194] Suitable habitat for microorganisms and protection from predators might be provided by the high porosity and substantial surface area of biochar.[181] Nutrients from the soil can be more effectively absorbed by biochar with a larger surface area and higher cation exchange capacity than they can be through leaching out of the soil.[181] While predatory mites, nematodes, and protozoans cannot access biochar because its pores may be less than 5 μm in diameter, bacteria and fungus hyphae can.[195] Oil palm kernel shell biochar generated industrially has a large surface area (135 m^2/g) and an average pore diameter of 4.93 nm, making it appropriate for NP fertilizer impregnation. The industrial oil palm kernel shell biochar contains nutrients (K, Mg, Ca) that are good for plant growth.[145] Palm kernel and rice husk gave pore volume (0.65 and 0.0003 cm^3/g) and pore diameter (16.92 and 24.06 Å), respectively.[151] Also, maize straw showed pore volume and diameter of 0.013 cm^3/g and 7.33 nm, respectively.[167] Corncob, rice straw, rice husk, and cow manure showed a pore volume of (0.012, 0.07, 0.211, 0.099 cm^3/g), and pore diameter (1.68, 3.78, 2.72, 3.75 nm), respectively.[168] A study also reported the pore volume (0.376, 0.019 cm^3/g) and pore diameter (3.826, 3.910 nm) of chitosan and kitchen waste biochar, respectively[171]

Functional groups

Because of its abundant surface functional groups (such as carbonyl, hydroxyl, phenolic hydroxyl, and carboxyl), large specific surface area, high pH, and high cation exchange capacity, biochar application aids in the immobilization of organic and inorganic contaminants.[124] The presence of peaks for hydroxyl, carboxyl, and alkyl in rice husk biochar and carboxyl in palm kernel biochar revealed that both biochar could enhance nutrient retention in soil.[151] Manure, rice husk, and corncob possess oxygen-containing functional groups that aid in the complexation of Cd, such as hydroxyl (-OH), carbonyl (-C=O), and ether (-C-O).[168] Wheat straw contains a functional group with a role in the metabolism of glycan biosynthesis and metabolism, energy, cofactors, vitamins, nucleotides, and amino acids.[178] The development of Pb and Cu complexes with organic functional groups found in the biochar might be responsible for the increase in the oxidizable proportion.[196,197] The ability of oxygen-containing functional groups on maize straw surfaces to bond with soil Al^{3+} reduces the toxicity of soil Al.[167] By creating stable compounds through the interchange of ligands with hydroxyl functional groups, biochar may lower the phyto-availability of trace

Table 6. Production processes and functions of various residue-derived biochar

Country	Biochar	Production process	Function	References
Peninsular Malaysia	Palm kernel shell, Rice husk	Pyrolyzed at 400°C Gasified at 800°C	influences the microbial community in the short-term amendment by changing the soil's habitat and adding more labile active carbon sources.	Simarani et al.[151]
Dubai UAE	Date palm leaves	Pyrolyzed at 350°C	helps maintain soil fertility by increasing microorganisms that stimulate plant development.	Khan et al.[162]
USA	Dairy manure, wood	–	generated an increase in soil prokaryotic microbial richness that was concentration-dependent and correlated, independent of crop, with higher soil functional capabilities.	Obayomi et al.[161]
China	Filtered mud from sugarcane stem	–	enhanced the slow-release of NO3- and the chemical characteristics of the soil, enhanced growth characteristics and elevated soil nutrient cycling enzyme activity, boosted the number of beneficial bacteria and the fungal responsible for breaking down the lignocellulose complex, while decreasing the diversity of soil bacteria and having little effect on fungus diversity.	Abubakar et al.[163]
China	reed straw, Cow manure	Slow pyrolysis	exhibited higher potentials for improving peanut development in a coastal salt-affected soil and dramatically increased fresh shoot weight.	Sun et al.[164]
China	Wood, Peanut shell	–	Peanut shell biochar dramatically decreased the variety of bacteria and fungi.	Ng et al.[131]
Poland	Relict charcoal hearths	pyrolysis	could significantly improve the microbial variety of forest soils, particularly when replanting or recultivating disturbed areas.	Lasota et al.[165]
Canada	Pine chips	Pyrolysis 700°C	used in conjunction with paper mill biosolid to increase soil microbial activity in clay soil; this could improve nutrient cycling without negatively affecting soil microbial communities.	Manirakiza et al.[166]
China	Grapefruit peel	pyrolysis	NH4+-N and Mn2+ pollution in soil and water can be greatly reduced with the use of strain AL-6.	Deng et al.[132]
Switzerland	Birch wood	Pyrolysis at 450°C and 700°C	With no negative impacts on soil microbial communities, they enhanced the uptake of vital elements such as Zn and K	Evangelou et al.[152]

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Table 6. *Continued*

Country	Biochar	Production process	Function	References
China	Maize straw	Pyrolysis at 400°C	Accelerate microbial metabolism to boost microbial activity and enhance acidic soil's general health	Xia et al.[167]
China	pig, duck, silkworm, cow manure, rice straw, rice husk, and corn cob	300, 500, 700°C	notable increases in the chemical fertility of the soil	Guo et al.[168]
China	Cattle/poultry manure	–	enhanced rice yields and soil health with a reduction in CH_4 emissions.	Iqbal et al.[169]
China	Rice straw (roots, stalks, leaves)	–	enhancing soil fertility and maintaining crop productivity	Liu et al.[170]
China	Pig manure, corn straw	Pyrolysis at 400°C	may modify soil characteristics, soil fertility and nutrient availability, and the fate and effects of pollutants to positively shape soil microbial communities in different ways.	Xiang et al.[75]
China	chitosan straw, raw biochar/ kitchen waste	–	enhance fertility, and immobilize heavy metal	Liu et al.[171]
China	Rice straw	Pyrolysis at 500°C	reduced α-diversity of the bacterial community but increased soil fertility, changed the bacterial community's structure, and enhanced the roles of soil bacteria.	Xu et al.[172]
China	Rice straw+slag	Pyrolysis at 600°C	increased SOC content	Lin et al.[130]
China	Straw return	–	enhanced bacterial and fungal community composition, enzymatic activity, soil fertility, and community diversity relative to conventional planting	Yang et al.[173]
China	Rice straw	–	decrease soil Fe bioavailability and effectively encapsulate heavy metals; nevertheless, soil Mn bioavailability is raised, potentially affecting soil micronutrient productivity and fertility.	Xu et al.[174]
China	Straw return	–	notably increased the contents of soil organic carbon, mineral N and accessible P	He et al.[175]
China	Wheat and maize straw	–	a beneficial way to increase crop productivity and soil properties	Cui et al.[176]
China	Moutai distillers grain biochar	Pyrolysis at 550°C	can lower the proportion of some pathogenic fungus and act as a barrier to stop dangerous plant diseases from growing and reproducing.	Zhang et al.[177]
China	Wheat-crop straw	Pyrolysis at 550°C–650°C	altered the structure of the fungal community and enhanced the diversity and quantity of soil fungus species.	Gao et al.[122]
China	Feedstock of wheat straw	Pyrolyzed at 500°C	enhance soil fertility and yield a few years after treatment.	Wang et al.[178]
China	Rice straw, pig manure	Anaerobic condition of 450°C	increase the yield of sweet potatoes and rapeseed while improving the quality of the soil in comparison to artificial fertilizers.	Li et al.[179]

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Table 6. *Continued*

Country	Biochar	Production process	Function	References
China	Rice straw	Slow pyrolysis at 500°C	In intensive vegetable cultivation, a 20 mg ha−1 biochar application rate in conjunction with N fertilization appeared to be recommended to produce the best vegetable output with the lowest N2O emissions.	Li et al.[165]
China	continuous straw return	Pyrolysis at 450–600°C	enhances soil sustainability	Bai et al.[180]
China	Rice straw biochar	Pyrolysis at 450°C–550°C	enhanced the root biomass, tiller number per unit area, and plant height maximum growth rate.	Huang et al.[157]
China	Wheat straw	350°C-550 °C at vertical kiln	potentially modify the makeup of microbial communities and boost wheat yield and abundance.	Liu et al.[181]
China	Reed, smooth cordgrass, rice straw	Pyrolysis at 450 ± 20°C	Compared to the matching non-pyrolyzed straw incorporation, every biochar incorporation reduced soil respiration.	TIAN et al.[182]
China	Rice straw	Pyrolysis at 500°C	Plant growth stage and the physicochemical characteristics of the amendment have an impact on the functional diversity of microorganisms in organic treated paddy soils.	Yuan et al.[183]
China	Rice straw	Pyrolyzed at 550°C	could enhance the microbial community structure and soil nutrient status, encouraging fungal community variety in soils used for tobacco cultivation.	Gao et al.[184]
China	Maize straw	Slow pyrolysis at 400°C	showed that the biochar greatly raised the quantities of organic carbon and nitrogen while maintaining the pH and lowering the amounts of mineral nitrogen and the microbial quotient.	Zhu et al.[128]
China	Straw and bamboo	Slow pyrolysis at 600°C	enhances soil fertility.	He et al.[185]
Denmark	Wheat straw gasification, fresh straw	Treated at 550°C	had a negligible impact on the structural and functional diversity of soil microbial communities in a temperate Danish agroecosystem.	Imparato et al.[186]
Spain	rice husk, rice straw, wood chips of apple tree, and oak tree	Pyrolyzed at 400, 500, 600, 700, and 800°C	–	Jindo et al.[160]
USA	Manure	Fast pyrolysis at 500°C	–	Elzobair et al.[187]
Poland	Apple, pear plum	Fast pyrolysis at 600°C, slow pyrolysis at 400°C	–	Kazimierski et al.[142]
Cyprus	biosolids-derived, cattle-derived pyrolyzed at 550°C	–	might be used to improve lettuce plants' development and, to some extent, their nutritional value.	Christou et al.[125]
China	Maize straw, rapeseed	Pyrolyzed at 550°C	decreased phytoavailability and uptake by maize, particularly in situations with high moisture content; redistribution of Cu and Pb among their more stable geochemical constituents; and reduced mobility.	Salam et al.[136]

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Table 6. *Continued*

Country	Biochar	Production process	Function	References
Pakistan	Wheat straw	Slow pyrolyzed at 300°C–350°C	–	Khan et al.[143]
Canada	wheat straw, timothy grass and pinewood	Pyrolyzed at 450°C	Slow heating rate biochars' extremely alkaline composition suggests that croplands could use them to improve soil fertility and crop productivity.	Mohanty et al.[144]
China	Rice straw	Fast pyrolysis at 600°C	–	Fan et al.[188]
Malaysia	Palm kernel	Slow pyrolysis at 400°C	–	Ajeng et al.[189]

elements.[136,198] It is important to note that as the pyrolysis temperature rises, the amount of functional groups on the biochar surface decreases.[199] More oxygen-containing functional groups were present in old biochar in soils than in fresh biochar on top of the soil.[185] The pore structure is mostly improved by physical activation, while chemical activation (using chemicals such as HNO_3, H_2SO_4, H_2O_2, KOH, $ZnCl_2$, and Na_2CO_3) raises additional functional groups in biochar while simultaneously increasing the specific surface area.[200] The addition of KOH influenced and made nitrogen and oxygen-containing functional groups more visible after KOH alteration..[201,202] Functional groups that were present in the biochars, such as rice straw (hydroxyl, carbonyl, carboxyl, amino group),[170] chitosan straw and biochar kitchen waste (hydroxyl, carboxyl, amino group),[171] rice straw + slag (hydroxyl, carbonyl, aldehyde, phenolic),[130] maize straw, rapeseed (hydroxyl, carboxyl, phenolic),[136] wheat straw, and timothy grass (carboxyl, hydroxyl, phenols, alcohol, nitrile, alkyne) are reported.[143,144]

Thermal stability

Due to its high chemical and thermal stability, biochar may not be the best source of carbon for microbes, but because of its porous structure, capacity to absorb nutrients, and ability to retain water, it may still be able to offer habitats and nutrients that have a significant impact on microbe growth and interactions with the environment.[128] Wheat straw showed stability to warming,[181] while aged biochar showed greater stability and lower negative charge content for maize straw.[167] The incorporation of silica elements into the chemical structure of rice-based biochar leads to a high yield and distinctive chemical properties.[160] A biochar yield of 32.8–59.6% was shown by poultry litter, sawdust, rice hull, and sugarcane with increasing pyrolysis temperature demonstrating the stability of biochar independent of the feedstock,[140] while eucalyptus wood-derived biochar resulted in a dramatic decrease of biochar with the temperature increase from 450°C to 650°C.[141]

EFFECT OF BIOCHAR ON THE PHYSICOCHEMICAL CHARACTERISTICS OF THE SOIL

Biochar is a carbon-rich stable material derived through anoxic or limited oxygenated thermal conversion of a wide-ranging biomass.[203] Due to the high stability within the soil matrix can last for thousands of years in the soil. The main purpose of adding biochar to soil is to sequester carbon and stimulate soil fertility by improving soil quality. Additionally, biochar-amendment soil served as a habitat for a variety of life forms (including useful bacteria and fungi), which significantly enhances soil microbial respiration. The physical properties of biochar act as a functional tool for the soil microbial ecosystem by creating space for soil microbes. Kumar et al.[58] reported that biochar significantly reduces soil hardening and density by increasing the soil cation-exchange capacity (CEC) and aeration and modifying the soil structure. In another study, Singh et al.[30] stated the potency of biochar to enhance crop yield by improving the soil properties, carbon content, and soil organic matter. It has been reported that biochar-amendment soil improves soil quality and improves yields due to the consistency and changes in

physical and chemical properties of soil, including soil pH, CEC, retained moisture, organic carbon, bulk density, and porosity.[30]

Influence of biochar on the chemical properties of soil

The chemical properties of soil designate the ability to offer a potential source of chemicals and nutrients to plants.[204] Intensive cropping and extensive use of chemical fertilizer resulted in a substantial decrease in soil organic matter, which triggered declining soil productivity. Soil organic carbon, an important index of soil quality, also decreased the structural stability of the soil. The addition of compost into the soil enhances the soil's organic carbon, stabilizes organic carbon, and improves the aggregate distribution in the soil matrix. Compost also increases the soil humic and semi-humic C source resulting in the long-term sequestration of organic carbon, soil organic matter, and NPK nutrients. Neemisha and Rani[29] reported the modification and stabilization of soil pH, electrical conductivity, and cation exchange capacity by applying compost to the soil matrix. The literature also suggested that the application of composts with high EC values resulted in enhanced soil organic carbon, mineral nutrition of plants, and increased microbial and enzyme metabolism.

Soil pH

Biochar-amendment soil shows significantly increased pH compared to the control (raw soil). Lehmann et al.[205] reported a substantial increment of soil pH after biochar addition, regardless of feedstock. Moreover, the literature also suggested a significant enhancement of soil pH for all types of soil textures.[206] However, in another study, Ding et al.[129] reported a noticeable increment in soil pH for the coarse-textured soils. The study also revealed that the coarse-textured soil matrix showed a greater response (positive increment) to the soil pH than the fine-textured soils. The increment in soil pH after biochar addition might be due to the alkaline nature of biochar and increasing soil nutrients (P, K, and S).[207] The study also showed that biochar-modified soil increased soil pH and soil organic matter and increased nutrient availability (for plant uptake) which significantly enhanced the crop productivity and growth of wheat. It is also documented that both the high and low pyrolyzed biochar significantly increased soil pH compared to the control.[129] Moreover, the rise in soil pH was further noticeable for manures-mediated biochar (64%)> wood-mediated biochar (58%)> waste lignocellulosic biochar (47%)> and herbaceous feedstocks-mediated biochar (26%).[30] In addition, no study was reported on size-dependent (particle size of the biochar) variations of pH for agricultural soil, and higher (49%) and lower (47%) rates of biochar modifications showed a noticeable enhancement of soil pH than the medium rate (34%) biochar amendment.[30,207]

Soil organic carbon

Biochar-modified soil shows a significant increase in soil organic carbon (OC) compared to the control soil. The amount or quantity of biochar was significantly positively correlated to the amount of OC, moreover, no substantial differences were observed with the biochar produced from different feedstocks or biomass.[129] However, Singh et al.[30] reported the maximum OM content was observed for wood feedstock-based biochar (30%) in compression with herbaceous (28%) and lignocellulosic (25%)waste biomass-based biochar, respectively. An outstanding improvement of OC has been reported for all types of soil texture, but the variations in the OM content among soil textures were insignificant.[208,209] Bista et al.[207] reported that soil P, K, and S concentrations were positively correlated with the OC, while soil NH_4-N and NO_3-N were negatively correlated with the biochar-modified soil. Moreover, pyrolysis temperature played a significant contribution to the quantity of OM in biochar.[80] Biochar prepared at low temperatures showed a higher amount of OM compared to the biochar produced at high temperatures, which might be due to the ash formation and decomposition of in the degasification of lignin, cellulose, hemicellulose, and other phytochemicals.[127,210–212] Bista et al.[207] reported that wheat biomass productivity was positively correlated with the OM present in the soil-biochar matrix.

Soil cation-exchange capacity

Soil CEC plays a critical role in nutrient transport in plants and crops. The biochar-modified soil matrix observed a significant increase in soil CEC compared to the control. The highest enhancement of CEC was observed for coarse and fine-textured soils (37% and 39%, respectively) (Bista et al., 2019). Both the high and low-temperature pyrolyzed biochar significantly increased the CEC, but the low-temperature pyrolyzed biochar showed a maximum increment of the CEC of the biochar-soil matrix compared to the control (Singh et al., 2022). The enhanced soil ECE with low thermal pyrolysis might be due to the retention of the phytochemicals, including lignin, cellulose, and hemicellulose.[203] In addition, the increment of soil CEC in the soil-biochar matrix was 22%, 37%, and 45% for lower, medium, and high application rates, respectively. However, no significant transformation of the soil-biochar matrix was observed for biochar prepared from different feedstocks.

Soil EC

The overall changes in soil EC were not found to be significant after biochar modification. There was a moderate increase of soil EC for fine-textured soils (55%) compared to the control. Jones et al.[213] carried out a field trial in the UK and found it to be an insignificant influence on the soil electrical conductivity (from 46 to 43 $\mu S\ cm^{-1}$) after three years of biochar addition. However, no significant changes were observed for the other soil textural groups (Lehmann et al., 2011). Singh et al.[30] reported that the lower temperature pyrolysis of biomass enhanced the soil EC compared to the lower temperature pyrolyzed of biochar. The study also revealed that the higher rates (>80 $t\ ha^{-1}$) of application of biochar significantly increased in soil EC, which was 43% greater than the control.

Influence of biochar on the physical properties of soil

Bulk density

Biochar modification generally reduces the bulk density of the soils-biochar matrix compared to controls. Githinji,[214] reported decreasing bulk density with increasing the quantity of biochar. The result showed that 25%, 50%, and 75% biochar amendment resulted in 1.09 $g\ cm^{-3}$, 0.89 $g\ cm^{-3}$, and 0.61 $g\ cm^{-3}$, respectively, compared with the bulk density of the control soil 1.33 $g\ cm^{-3}$. A study by Singh et al.[30] suggested that a biochar-treated soil matrix (synthesized from different feedstock) significantly reduced the bulk density compared to the control. The study also suggested the highest reduction of the bulk density was observed with coarse soil-textured matrix (36%)

followed by fine-textured soil matrix (19%) and medium-textured soils (14%) matrix. However, the effects of animal manure-based biochar on bulk density were not reported due to the limited experimental data. In addition, there were no significant changes observed for the biochar produced at high (31%) and low (28%) pyrolytic temperatures in comparison with the control soil.[207] Literature also suggested that the reduction in bulk density was meaningly greater (56%) for the higher rate of biochar modification followed by the lower rate of biochar modification (45%), where the lower rate of biochar modification showed the most downward decline (18%) bulk density.[124]

Soil porosity

Generally, soil porosity is indirectly proportional to the bulk density and was significantly increased with biochar modification compared to the controls. The enhanced magnitude of incremental soil porosity was observed for all feedstock types and all types of soil textures compared to the control. Studies showed biochar produced by both the high and low pyrolytic temperatures significantly enhanced the soil porosity, but biochar produced at higher temperatures showed higher porosity (66%) than the biochar produced at a lower temperature (56%).[30] Bista et al. [208] reported increasing rate of biochar amendment boosted the soil porosity. The study showed the highest Soil porosity (78%) with high application rates, followed by a lower rate of biochar application (52%).

EFFECT OF BIOCHAR ON THE MICROBIAL AND FUNCTIONAL BIOLOGICAL CHARACTERIZATION OF THE SOIL

The intricate biomass on Earth comprises one billion microorganisms per gram of soil and includes bacteria, fungi, algae, protozoa, and nematodes.[215] These microbes play a significant role in shaping the diversity of plant communities and are crucial for sustaining the essential functioning of terrestrial ecosystems. The influence of various biotic and abiotic factors is primarily mediated through microbial population diversity measured by biomass, species richness, evenness, and community composition and their extracellular enzymes which regulate processes such as nutrient cycling, carbon and nitrogen fixation, and organic matter decomposition. Consequently, soil health and fertility, along with biodiversity, emerge as vital indicators.[28,216,217] Microbes are particularly vulnerable to diverse environmental stressors, including heavy metal contamination, biogeographical factors, improper fertilizer application, drought, and temperature variations.[218] This underscores the significant challenges in maintaining soil fertility and mitigating the reduction of beneficial microbial diversity within the agriculture ecosystem. A recently proposed solution involves the incorporation of biochar, which aims to enhance soil fertility, reduce emissions, and positively impact soil microbial diversity and activity.[219] The application of biochar-based fertilizers leads to nutrient recovery and the maintenance of microbial populations, making them highly valuable organic fertilizers with notable steadiness and nutrient-holding capacity. Additionally, biochar incorporation augments enzymatic activity, exerting a significant impact on the availability of nutrients to plants, consequently impacting overall plant productivity. Assessment of soil enzymes proves effective in appraising soil quality, owing to their remarkable sensitivity and rapid responsiveness to change the soil milieu.[220]

Researchers have placed significant emphasis on the reduction of heavy metal contamination in soil and water, resulting in prompted responses from microbial populations. Diverse biochar variants, sourced from materials such as chicken dung, sulfur-modified rice husk, wheat straw, sewage sludge, casuarina, and corn straw are generated from a spectrum of biomass categories such as agriculture, forestry, industry, and municipal solid waste. Biochar products can adsorb and fix organic contaminants, positively impacting the extractability of heavy metals such as Cu, Hg, Cd, Co, Cr, Cu, Ni, Pb, and Zn in water and soil, as well as enzyme activity such as urease, reductase, phosphatase, and catalase in various agricultural contaminant lands.[184,221] In contrast, the short-term application of biochar amendment had no significant effect on microbial community structure when compared with manure amendment.[187] However, recent reports suggest significant changes in richness and diversity observed with the addition of biochar consisting of leaf and woodchip.[173,222] Additionally, the application of biochar shows a notable impact on microbial diversity within both soil and plant rhizospheres. Studies suggest that the amendment of biochar in various contaminated soils exposed to heavy metals, fertilizers, and pesticides serves to protect specific microbial populations, thereby benefiting plant growth and development.[223] While extensive literature is available on the impact of biochar on microbial communities, the interaction between biochar and microbes remains a subject of debate and is yet to be fully elucidated. For instance, the porous structure of biochar, characterized by micro (<2 μm) and meso (2–50 μm) pores, provides a habitat for microbes, facilitated by structures such as flagella, pili, and fimbriae that act as habitats ultimately contributing to enhanced soil fertility.[224]

To dissect and quantify the diversity of the microbial population in biochar-based fertilizers, researchers have used a variety of techniques, including colorimetric and fluorescent measurement, traditional culturing techniques, hyphal abundance, fatty acid analysis, and restriction fragment length polymorphisms (RFLP).[225,226] However, traditional microbial culture techniques have limitations in understanding the low number of organisms that can be cultivated in the laboratory and molecular functional characterization of plant-microbes-biochar interactions (Figure 5). Significant developments in next-generation sequencing technology have created new opportunities for the biotechnological application of this technology, ranging from personalized medicine to soil health, by providing new insights into the molecular groundworks of microbial ecology. Firstly, the community taxonomic composition (operational taxonomic units or OTUs) and phylogenetic diversity of the sample without culturing are frequently assessed by amplicon DNA sequencing using markers that include the internal transcribed spacer region (ITS) of fungi and the 16S ribosomal RNA subunit of bacteria and archaea[227,228] (Table 7; Figure 5), and second, utilizing the shotgun metagenomics sequencing provides both diversity and functional characterization (Table 7) of the microbial population which offers a wide range of applications that meet requirements for plant growth and development.[224] A significant

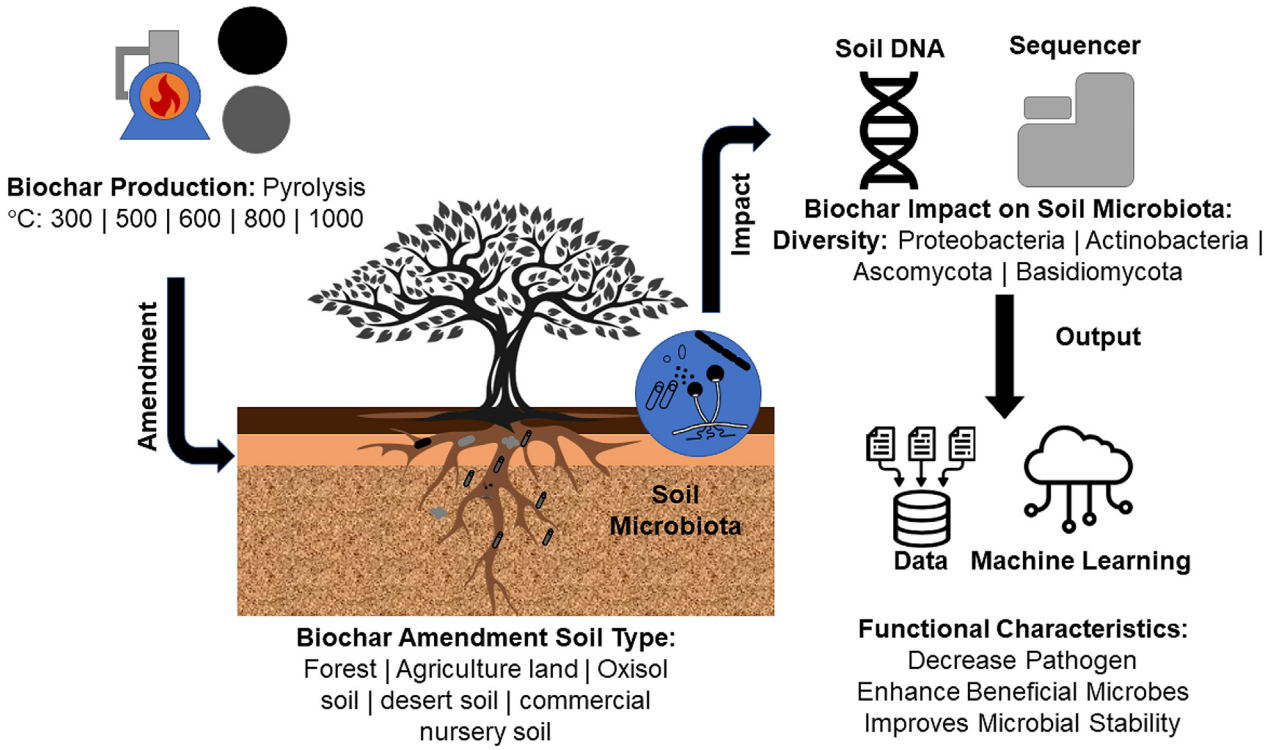


Figure 5. Biochar impact on soil microbes and recent advancement of molecular tools and bioinformatics analysis to dissect microbial population and their functional characteristics

obstacle that biology researchers face right away is bioinformatics analysis of the raw sequence data that is obtained from the sequencer. The 16S rRNA gene is relatively conserved, although a small number of base pairs can change in a very short amount of evolutionary time due to hypervariability in some of the sequenced areas (V1 – V9). Most studies that focus on the hypervariable region of the 16S rRNA gene (V3-V4 and V4 – V5) simultaneously cover the bacterial and archaeal domains. Sequence assembly and stitching together individual DNA sequences is a crucial stage in the analysis process. Metagenome-assembled genomes (MAGs) are produced through binning assembled contigs, that have comparable properties and undergo quality filtering. As a result, MAGs might be a representation of a microbial genome of highly comparable microbial strains.[227,237]

According to earlier research using the amplicon and shotgun metagenome sequencing techniques on biochar amendment in a variety of soil types including forestry and agricultural lands, Proteobacteria, Bacteroidetes, Chloroflexi, Firmicutes, and Actinobacteria have relative abundances, and dominant Table 7. Actinobacteria is a common bioremediation category, especially when it comes to breaking down polycyclic aromatic hydrocarbon (PAH)- in the soil. However, the relative abundance of Actinobacteria was shown to have dropped in Biochar-layer-by-layer samples, whereas Proteobacteria and Bacteroidetes significantly outnumbered them.[232] In the variety of biochar amendment samples, the distribution of bacteria at the genus level including *Acidobacter*, *Gemmatimonas*, *Gemmatirosa*, *Micromonospora*, *Ramilbacter*, *Rhizobiales, Sphigomonas*, and *Streptomyces* significantly fluctuated Table 7. According to Trivedi et al.,[238] *Acidobacter*, for instance, is known to be slow-growing, or "oligotrophs," with high C consumption efficiency and an abundance of refractory organic materials in the soil. In a similar vein, it has been observed that *Gemmatimonas* reduce their relative abundance following the addition of fresh organic matter and that they are more likely to degrade the organic C already present in the soil than to add new organic matter.[239,240] Proteobacteria presence is influenced by crop selection and nitrogen content. Additionally, the rhizosphere sample of nitrogen fertilizers contained a significant and abundant class of alphaproteobacteria and the group led by Wang et al.[231] was able to observe the same class dominating in soil supplemented with biochar. It is reported that members of this class, such as *Sphingomonas* and *Rhizobales,* are involved in the inorganic compounds' breakdown or degradation. Enzymes that metabolize plant-associated carbohydrates such as cellulose and hemicellulose are crucial for the metabolism of bacteria associated with roots. Dror et al.[235] discovered that MAG2, which is binned to *Lentzea* spp., carries many enzymatic genes involved in cellulose degradation. It is well known that various Actinobacteria groups (primarily *Streptomyces* and *Micromonospora* spp.) can break down chitin.

According to recent results suggest that next-generation sequencing helps scientists to understand the root rhizosphere dynamics under biochar amendment and the functional characteristic that inhibits the growth of fungal pathogens.[231,234,241]

Table 7. Summary of various biochar impacts on microbial diversity analysis and functional biochemical characterization using high-throughput sequencing and their applications

Biochar varieties/ Study area	Geographical location/study conducted year	Biochar Production Process (Temperature)	Application	Sequencing Method	Phylum level Abundance (Bacteria and Fungi)	Functional Characterization	Key Findings	References
Forest biochar/ Sugar maple and wood chips	Canada/fall 2010	Pyrolysis (500°C reached 800°C)	Terrestrial ecoregions and between upland and lowland environments/ High carbon, minerals	Amplicon and shotgun metagenomics	Dominance of Proteobacteria, and Actinobacteria. Increased diversity of Basidiomycota	Nitrite reductase increased. CO_2 fixation genes	Forest biochar promoted diverse and functionally rich microbial communities. Enhanced soil fertility	Noyce et al.[229]
Mixed feedstock biochar	Hawaii, USA	Pyrolysis (600°C) produced by Diacarbon Inc	Oxisol soil Napier grass cultivation/ High surface area	DNA stable isotope probing with shotgun metagenomics	Increased Proteobacteria, Actinobacteria, Ascomycota and Basidomycota. Metagenome-assembled genome of Nitrospira and Bradyrhizobium	Nitrite reductase Denitrification genes (*nirK* or *narG*)	Resilience of active microbial communities	Yu et al.[230]
Cotton straw-derived biochar/China	China/2014 - 2018	Pyrolysis (450°C) produced by Diacarbon Inc	Soil nitrogen transformation in a drip-irrigated cotton field	Shotgun metagenomics	Enhanced Nitrosomonas, Nitrospira and reduced abundance of soil fungi	Increased soil nitrogen transformation	Biochar enhanced soil nitrogen transformation processes by affecting microbial community composition	Wang et al.[231]
Mixed biochar hybrid modified bio-microcapsules	China	Pyrolysis (1000°C) produced by Hainuo Charcoal Company, China	Enhance phenanthrene degradation	Shotgun metagenomics	Proteobacteria and Actinobacteria	Amino acid and carbohydrate metabolisms dominate during the remediation process	Biochar hybrid modified biomicrocapsules significantly enhanced phenathrene degradation by altering microbial community structure	Deng et al.[232]

(*Continued on next page*)

Table 7. ***Continued***

Biochar varieties/ Study area	Geographical location/study conducted year	Biochar Production Process (Temperature)	Application	Sequencing Method	Phylum level Abundance (Bacteria and Fungi)	Functional Characterization	Key Findings	References
Powder bamboo biochar	China	Slow pyrolysis (800°C)	Aerobic composting	Shotgun metagenomics	Firmicutes, Proteobacteria, and Actinobacteria	Significant impact on nitrogen metabolism (*nirK* and *nosZ*) and inhibited glutamate dehydrogenase activity	Powder bamboo biochar effectively reduced N_2O gas emissions during composting.	He et al.[233]
Pig manure wheat straw Biochar	China	Pyrolysis	Degradation of antibiotics resistant gene - aerobic composting	Shotgun metagenomics	Firmicutes and Actinobacteria	Biochar decreased the relative abundance of *tetW* and *tetL*	Biochar effectively reduced antibiotics and antibiotics resistance genes in pig manure composting. Enhanced beneficial microbial communities.	He et al.[234]
1% eucalyptus Biochar/ High carbon, essential nutrients	Israel	Pyrolysis (600°C)	*Fusarium oxysporum* pathogen inhibition	Amplicon and shotgun metagenomic sequencing	Enhanced Bacillus, Pseudomonas, reduced *Fusarium oxysporum*	Carbohydrate metabolites involved in cellulose degradation	Biochar facilitated antagonism against Fusarium oxysporum through microbial community changes.	Dror et al.[235]
Sewage sludge biochar	India	Pyrolysis (300°C–600°C)	Mitigation of solid waste	Amplicon sequencing	Proteobacteria, Actinobacteria, Acidobacteria, increased soil saprophytic fungi	Enhanced soil nutrients, plant biomass	Sewage sludge biochar improved soil nutrient status, microbial abundance, and plant biomass, providing a sustainable waste management approach.	Ahmad et al.[236]

However, the conditions and mechanisms of microbial dynamics and immobilization by biochar remain unclear. Therefore, many investigations are necessary to conduct in-depth research on microbial community characteristics combined with multi-omics (genomics, metagenomics, metatranscriptomics, and metabolomics). Additionally, the process of assembling and binning shotgun metagenomic sequences has allowed the recovery of genomes from both abundant and rare microbial populations, known as metagenomic assembled genomes or MAG across diverse environments influenced by biochar soil amendment (Table 7). Furthermore, these metagenomics techniques are helping to uncover novel functional signatures and adaptive traits of the microbial taxa, bridging the gap in our understanding of the connection between microbes with ecological processes.[230] Biochar addition can result in positively and negatively regulated sequestration of C, carbohydrate metabolism, nitrogen metabolism, and denitrification are evidenced by the metagenomics investigation in several systems (Table 1). Primarily, the presence of genes responsible for chitin binding and degradation, as well as novel biosynthetic clusters (BGCs) were observed when fungal pathogen *Fusarium oxysporum f.* sp. *Radicis cucumerium* was introduced into the cucumber plant in both biochar-amended or nonamended soil. These findings suggest that the addition of biochar creates a "buffering effect" that mitigates the destabilizing impact of *F. oxysporum* f. sp. *radicis-cucumerinum* on the root-associated microbiome. This, in turn, helps to preserve beneficial microorganisms that produce enzymes and secondary metabolites that contribute to the overall plant health.[235] Antibiotic microbial resistance is a major health threat across the globe and antibiotics are massive contaminants of soil. A study investigates biochar-mediated antibiotics (oxytetracycline, chlortetracycline, and tetracycline) degradation during the aerobic composting of pig manure. The results show that biochar promotes antibiotic degradation and decreases the relative abundance of antibiotic-resistant genes (ARGs).[234] Bacterial immobilization was utilized in the bioremediation process and a group was involved in developing a hybrid modification of biochar with layer-by-layer (LBL) assembly microcapsules for the phenanthrene treatment. The study suggests that the LBL bio-microcapsule promotes bacterial community structure and improves functional metabolic genes.[232] The outcome of the Overall, biochar amendment has been proven to improve soil fertility and microbial population dynamics using various methodologies under multiple environmental conditions. Future research would be required and the integration of machine learning approaches which comprise data on biochar production, amendment soil type, time line, and microbial composition helps the researchers to understand the interaction with physicochemical properties, molecular biological mechanisms on the soil, and their optimization.

INSIGHTS INTO BIOCHAR-PLANT INTERACTION

The impact of continuously applying biochar over a decade on *Vitis vinifera*'s yield and fine root characteristics was investigated by Baronti et al..[242] Biochar treatment enhanced the root length of the upper soil and the deeper layer. Védère et al.[243] compared the effects of fresh and soil-aged biochar/compost blend on plant growth. Plants raised with aged blends showed a significant reduction in stress level, as evidenced by the proline levels, with a maximum of 5-fold reduction in plants raised in aged biochar/compost mixture.[243] Higashikawa et al.,[244] explored the effects of consecutive treatments of progressively higher biochar quantities on onion nutrition, crop yield, and soil fertility in a Humic Cambisol under a no-tillage system in Santa Catarina, Brazil, over two crop seasons. Notably, onion production was significantly boosted when biochar was combined with mineral fertilizers in both growing seasons. The combined use of timber waste biochar as a soil amendment and exogenous brassinosteroids as an aerial leaf spray to mitigate stress in wheat plants under deficit irrigation was investigated. Combined brassinosteroid and biochar treatment led to substantial enhancements in plant growth, mainly when a higher dose of brassinosteroid (3 μmol) was applied; this also led to a significantly increased root and shoot fresh weights, dry weights, and lengths of wheat plants.[245]

Snake weed treated with biochar enclosed in food waste digestate displayed considerable enhancement, indicating an augmentation in chlorophyll content, fresh weight, leaf area, and blossom frequency.[246] Biochar derived from a prevalent hemiparasite plant, mistletoe (Loranthus europaeus), was applied to oak seedlings. Numerous traits connected to ecophysiology, morphology, and growth were scrutinized. Seedling growth and ecophysiological responses benefited from moderate biochar concentrations, while the highest concentration produced unfavorable outcomes.[247] Pyrogenic carbon-derived biochar can alleviate the allelopathic effects of black walnut (*Juglans nigra*) and Norway maple (*Acer platanoides*). There was an approximately 35% enhancement in the total biomass of silver maple when leaf litter was treated with this biochar. In certain instances, it resulted in more than a 2-fold increase in the biomass of paper birch.[248]

Reduced urea application and the addition of biochar caused notable changes in cotton plant root system architecture, key enzyme activities in nitrogen assimilation, and cytokinin levels during the seedling phase. Decreasing urea by 1/4th and incorporating biochar into the surface soil layer increased active root absorption area, fresh weight, and nitrogen agronomic efficiency. Moreover, crucial genes such as amt1 (Ammonium transporter1), gdh3 (glutamate dehydrogenase 3), and gdh2 (glutamate dehydrogenase 2) showed substantial increases in very fine roots under conditions of reduced urea application combined with biochar incorporation.[249] Biochar-based slow-releasing fertilizers (BSRF), which include pyrolyzed cow dung, vermicompost, Lantana weed, and vermicompost leachate, have been shown to boost the shoot and root growth markedly and have a positive influence on the overall yield of Mung bean (*Vigna radiata*) in pot trials.[250]

A granular biochar-mineral urea composite was created by incorporating urea with biochar derived from green waste and enriched with clay minerals such as bentonite and sepiolite. Employing this Bio-MUC mixture resulted in a statistically significant enhancement in maize growth and an elevation in plant nitrogen content ($p < 0.05$). More precisely, there was a 13.8% rise in fresh shoot biomass and a 25.1% increase in fresh root biomass.[251] Frimpong et al.,[252] emphasized the potential synergies of integrating biochar with organic and mineral amendments in agriculture. Applying biochar alone in the first and second seasons did not significantly boost

maize and okra yields. However, a noteworthy increase in cassava yield occurred in the third season, likely attributed to the substantial release of nutrients from enhanced biochar decomposition during this period. The subsequent rise in maize, okra, and cassava yields when biochar was applied alongside compost and/or NPK fertilizer can be attributed to an additive effect, contributing to an overall enhancement of soil properties. The impact of granulated and standard biochars, featuring different particle sizes, on the performance of the resilient plant *Agastache foeniculum* was investigated. The findings demonstrated a significant improvement in plant growth linked to applying granulated biochar.[253]

Wood biochar derived from *Morus alba* L. and two beneficial soil bacteria, *Pseudomonas* sp., and *Staphylococcus haemolyticus*, effectively reduced drought stress for *Brassica napus* L. plants. When used alone or with biochar, they significantly improved canola plant growth, with *Staphylococcus haemolyticus* showing better results than *Pseudomonas* sp., even in stressful conditions.[254] Amaranth, alfalfa, and maize plants grown on soil amended with biochar exhibited differences in root morphology and growth when subjected to salinity stress compared to plants grown in soil without amendment. In addition to enhancing the photosynthetic pigments, soil enzyme activities (catalase, invertase, and urease) were also heightened, resulting in better nutrient accessibility.[255] Eucalyptus wood biochar was derived through pyrolysis, applied to twenty-day-old thyme (*Thymus vulgaris*) seedlings, and subjected to heat stress. The use of β-sitosterol and eucalyptus biochar markedly displayed a significant decrease in reactive oxygen species production and increased antioxidant synthesis with decreased cell electrolytic leakage. The results suggest that applying β-sitosterol and eucalyptus biochar mitigates heat stress effects by ensuring consistent availability of vital nutrients, enhancing resilience by elevating the synthesis of photosynthetic pigments, and improving antioxidant activity.[256] The salt tolerance of the banana varieties Williams and Grand Nain were enhanced when treated with silicon rich biochar with the increased production of osmoprotectants including phenols and proline.[257]

Peanut shell biochar or wheat straw biochar addition to soil containing *Ralstonia solanacearum* provided resistance to tomato plants compared to control. This is due to the reduced levels of Ralstonia along with increased microbial diversity in the biochar-amended soil.[258] Apple replant disease can cause drastic yield losses in apple orchards. Treatment of the *Fusarium oxysporum*-contaminated apple replant soil with biochar enriched the microbial diversity, increased the root indexes, and decreased the abundance of *Fusarium oxysporum*.[259] A comprehensive review of the effects of biochar on disease suppression in plants is provided by Iacomino et al..[260] Solaiman et al.[261] explored the link between biochar attributes and arbuscular mycorrhizal (AM) fungi in agricultural soil. Their findings revealed an inverse relationship between phosphorus concentration in biochar and AM colonization. Applying various biochars produced positive growth responses for wheat and clover. Particularly noteworthy were the highest levels of mycorrhizal root colonization, observed in subterranean clover with biochars from oil mallee and wheat with biochars from oil mallee, wood waste, and sawdust. Overall, biochar amendment to soil can improve the yield, enhance nutrient availability, help the plant in combating biotic and abiotic stress, and increase the microbial diversity in the rhizosphere (Figure 6).

CONCLUSIONS AND FUTURE PERSPECTIVES

In this review, we have focused on summarizing different aspects of biochar production techniques and diverse approaches of biochar-based soil rejuvenation including insights into biochar-plant interaction. The review reveals the traditional thermochemical processes such as pyrolysis and gasification, widely used for biochar production. However, it has been observed that the hydrothermal carbonization and torrefaction are gaining widespread usage as the properties of biochar obtained tend to be very different. These newer processes run at relatively lower temperatures and the biochar product possesses better heating value, oxygen functional, carbon content, and volatile matter. However, the surface areas were found to be lower compared to those from pyrolysis and gasification. Literature suggests the heating rate influences the quality of biochar as high heating rates favor the evolution of fluid phase products (gases and liquids), while slower heating rates may lead to longer residence times and higher energy consumption. Gradual changes in temperature can cause biochar materials to exhibit different behaviors, such as increased aromatization, surface area, and pore volume with rising temperature. While the feedstock type has a natural effect on the biochar production process, affecting hydrophilicity, aromaticity, mineral content, and porous structure.

Biochar is rich in oxygen-containing surface functional groups such as carbonyl, hydroxyl, phenolic hydroxyl, and carboxyl. These groups play a role in immobilizing organic and inorganic contaminants in the soil. It has been reported that the surface area and porosity contribute to its ability to enhance soil microbial adhesion. This, in turn, influences soil health and nutrient availability. However, some studies also showed no significant impact on microbial community structure with short-term biochar application, others indicate changes in richness and diversity with specific biochar compositions, such as leaf and woodchip. With regards to biochar-plant interactions, plants raised with aged biochar/compost blends show a significant reduction in stress levels, compared to plants raised with fresh blends. Consecutive treatments of progressively higher biochar quantities significantly boost onion production when combined with mineral fertilizers. Biochar derived from pyrogenic carbon alleviates the allelopathic effects of black walnut and Norway maple, resulting in enhanced total biomass of silver maple and paper birch.

Biochar-amended soil improves plant root morphology and growth under salinity stress, along with enhanced photosynthetic pigments and soil enzyme activities. Based on many of such related findings on biochar-plant interactions, biochar application can have diverse positive effects on plant growth, yield, stress resistance, and soil health across different crops and conditions. Biochar may be proposed as a solution to enhance soil fertility, reduce emissions, and positively impact soil microbial diversity and activity.

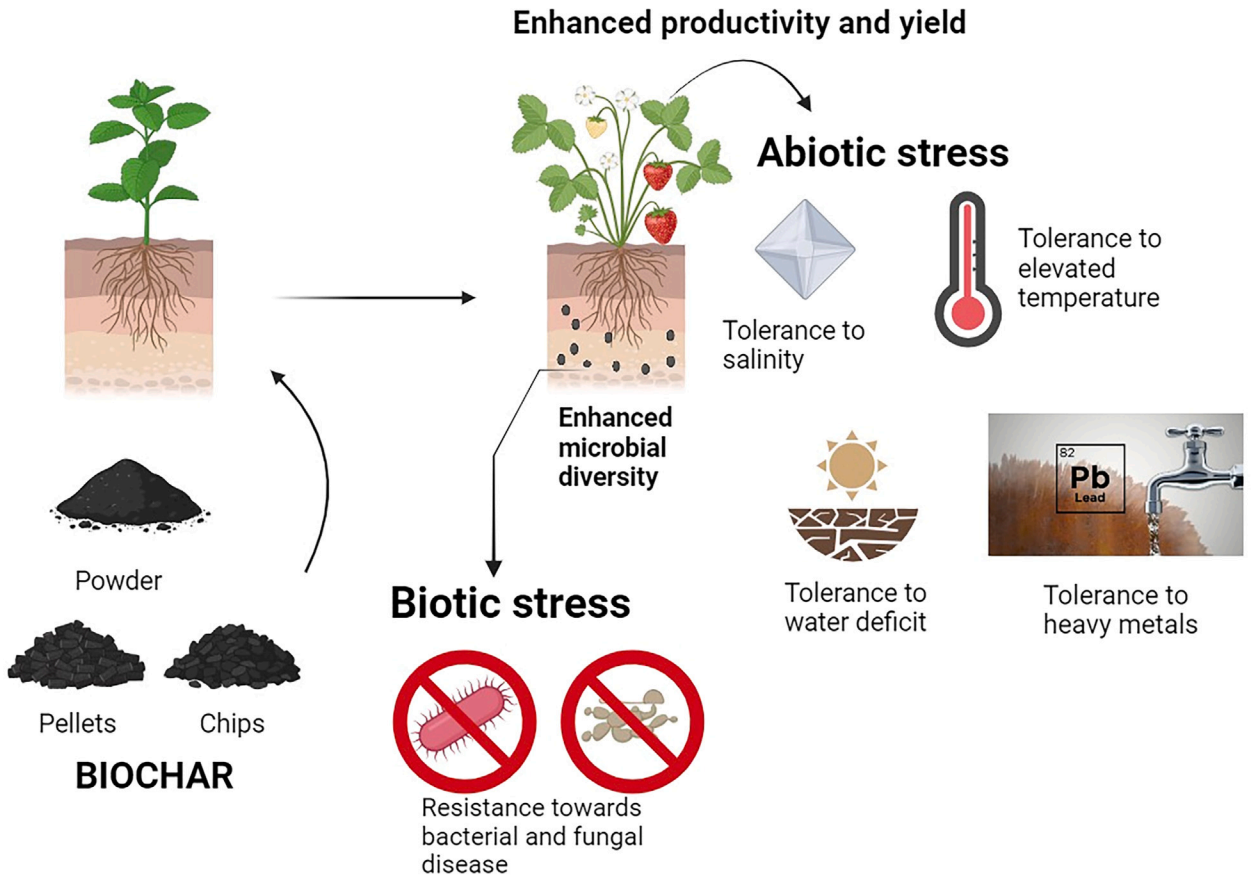


Figure 6. Illustration of the significance of biochar in soil amendment

Limitations of the study

Generally, the biochar properties vary significantly with production techniques, feedstock, and operating parameters. The short-term nature of many studies limits the understanding of long-term impacts on soil and microbial health under real life situations. In addition, environmental factors and potential contamination also pose risks, as does the economic feasibility of scaling biochar applications, especially with advanced production methods such as hydrothermal carbonization. These limitations underscore the need for comprehensive, long-term research to refine biochar's applications in agriculture.

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AUTHOR CONTRIBUTIONS

Conceptualization: JKB and RS; methodology: JKB and RS; investigation: JKB, WAS, and RS; validation: JKB and RS; writing – original draft: JOI, WAS, JKB, GBMM, SAC, and RS; writing –review and editing: JOI, CRO, VEO, MO, WAS, JKB, CSS, GBMM, SAC, and RS; supervision: JKB and RS.

DECLARATION OF INTERESTS

The authors declare that Prof. Jayanta Kumar Biswas is a guest editor for the special issue, "Biochar for Sustainable Environmental Management," but was not involved in the editorial handling of this review article.

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